

# Inverse $q$ -Analogues in All Parameters

Local Reversion, Singular Branches, Endpoint and Cyclotomic Asymptotics,  
Parameter Identifiability, and the Fabius–Rvachev Connection

A research extension of the ProveIt exponents-and- $q$ -series drafts

August 30, 2026

# Abstract

This report develops a parameter-wise inverse theory for finite and infinite  $q$ -Pochhammer symbols, Gaussian and multinomial coefficients,  $q$ -numbers,  $q$ -factorials,  $q$ -gamma,  $q$ -digamma,  $q$ -beta, Euler  $q$ -exponentials, basic hypergeometric series, theta functions, and a geometrically weighted family containing the Rvachev up-function and Fabius function. Every inverse is indexed by the parameter being recovered, a real interval or complex sheet, and a normalization path. This avoids the common but serious ambiguity of speaking about “the inverse” of a multiparameter special function.

The analysis is organized around a universal reversion calculus. Regular branches are computed to arbitrary order by Bell-polynomial recurrences, logarithmic coordinates, and multivariate Lagrange–Good inversion. Critical branches are treated by Newton–Puiseux series. Endpoint inverses are obtained by asymptotic reversion, including fractional powers, logarithms, Lambert  $W$ , periodic corrections, and root-of-unity radial transseries.

Several concrete results sharpen the earlier ProveIt inverse report. For the finite product  $F_n(a, q) = (a; q)_n$ , the complete nonconstant critical locus of the  $q$ -slice at  $q = -1$  is

$$a_*(2m) = \frac{1}{2m-1}, \quad a_*(2m+1) = -\frac{1}{2m+1}.$$

All these slices have an ordinary quadratic fold except

$$(a; q)_3|_{a=-1/3} = \frac{32}{27} + \frac{4}{27}(q+1)^3,$$

which has an exact cubic inverse. In the full  $(a, q)$ -plane this exceptional point is nevertheless a nondegenerate Morse saddle; the cubic occurs because the fixed- $a$  line is tangent to a null direction of the quadratic form. At a primitive  $m$ -th root  $\zeta$ , colliding zeros in the argument have multiplicity equal to the number of indices in one congruence class modulo  $m$ , giving explicit Puiseux exponents.

For the infinite product on a radial path  $q = \zeta e^{-t}$ ,  $\zeta^m = 1$ ,  $|a| < 1$ , we derive

$$\log(a; \zeta e^{-t})_\infty = -\frac{\text{Li}_2(a^m)}{m^2 t} + C_\zeta(a) - \mathcal{D}_\zeta(a)t + O(t^2),$$

with explicit convergent series for  $C_\zeta$  and  $\mathcal{D}_\zeta$ , and revert it to

$$t = \frac{A}{X} + \frac{\mathcal{D}_\zeta A^2}{X^3} + O(X^{-4}), \quad A = \frac{\text{Li}_2(a^m)}{m^2}, \quad X = -\log y + C_\zeta(a).$$

The cases  $q \rightarrow 1$  and  $q \rightarrow -1$  receive higher-order real formulas. A second, nonuniform regime  $a = q^x$  produces logarithmic corrections and a Lambert- $W$  preconditioner.

For Gaussian coefficients, coefficient positivity gives the global positive base inverse, reciprocal symmetry gives the  $q = \infty$  inverse, and palindromicity turns the expansion at  $q = 1$  into cumulant reversion for the partition-area distribution. At roots of unity, the cyclotomic valuation is exactly a carry indicator and therefore every forced zero is simple. The continuous lower-parameter inverse has two sheets meeting in a square-root fold at the symmetry center. Similar formulas are given for  $q$ -multinomials and for every positive palindromic  $q$ -polynomial.

For  $\Gamma_q(x)$ , the report separates the fractional-power Newton polygon at  $q = 0$ , the ordinary reversion at  $q = 1$ , and the reciprocal  $q = \infty$  regime. In particular,

$$\log \frac{\Gamma_{e^{-t}}(x)}{\Gamma(x)} = -\frac{(x-1)(x-2)}{4}t + \frac{(x-1)(x-2)(2x+3)}{144}t^2 + O(t^3).$$

The corresponding  $q$ -beta expansion has a critical curve  $xy = 1$ , on which base inversion changes from an ordinary Taylor branch to a square-root branch. The  $q$ -digamma base inverse has an analogous critical value  $x = 3/2$ .

Finally, the random series

$$X_q = (1-q) \sum_{j \geq 0} q^j U_j, \quad U_j \sim \text{Unif}[-1, 1],$$

contains the Rvachev law at  $q = 1/2$ . Its variance recovers the deformation parameter exactly:

$$\text{Var } X_q = \frac{1-q}{3(1+q)}, \quad q = \frac{1-3 \text{Var } X_q}{1+3 \text{Var } X_q}.$$

All even cumulants give independent monotone inverse observables. The accompanying Python program regenerates the symbolic tables, numerical checks, and figures at high precision.

# Relation to the source drafts and status of claims

The starting material is the collection of  $\text{\LaTeX}$  drafts under `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/ exponents-and-q-series` in the ProveIt repository, at the repository state consulted on August 30, 2026. In particular, the large forward monograph [1], the expansion report [2], the first inverse report [3], and the merged frontier survey [4] supply identities, conventions, and prior branch calculations. The present document is written as an extension: it rederives enough forward theory to make proofs readable, but concentrates on items that were absent or only briefly indicated there—multivariate inverse jets, parameter identifiability, the full  $q = -1$  critical classification, root-of-unity inverse normal forms, continuous upper/lower parameter inverses, and the generalized Fabius–Rvachev recovery problem.

Claims are tagged as follows.

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|                    |                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------|
| PROVED             | proved in this report, or reduced explicitly to a standard theorem;                                |
| SYMBOLIC/NUMERICAL | exact computer algebra or high-precision numerics reproduced by the archive;                       |
| CONJECTURAL        | a precise frontier statement supported by structural or numerical evidence, with no proof claimed. |

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Some formulas appear not to be standard in the exact form given here, but no claim of bibliographic priority is made. Radial root-of-unity asymptotics sit in a substantial literature on quantum dilogarithms and Nahm sums [12, 13]; recent gamma-product formulas for  $(z; q)_\infty$  provide another promising route to uniform inverses [14]. The distinction between a proved identity and a priority claim is maintained throughout.

# Notation and branch conventions

We use

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j), \quad |q| < 1,$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, \quad [x]_q = \frac{1 - q^x}{1 - q}.$$

Unless stated otherwise,  $n, k$  are integers with  $0 \leq k \leq n$ , while  $a, q, x, y$  may be complex. A symbol such as  $F_{q, \gamma}^{-1}$  means the inverse with respect to  $q$  on the sheet  $\gamma$ ; it never means a simultaneous inverse in all arguments. Complex powers and logarithms carry an explicit or contextually fixed branch. Radial limits use

$$q = \zeta e^{-t}, \quad \Re t > 0,$$

and continuation from positive real  $t$ . For a primitive  $m$ -th root  $\zeta$ , this convention fixes the branches in  $C_\zeta(a)$  and in the inverse transseries.

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## **Part I**

# **A general calculus of parameter-wise inversion**

# Chapter 1

## Inverse problems, sheets, and identifiability

### 1.1 One scalar equation does not recover every parameter

Let

$$F(p_1, \dots, p_d) = y.$$

For  $d > 1$ , this is generically a hypersurface of dimension  $d - 1$ , not a single point. A well-posed inverse problem must therefore do at least one of the following:

- (i) fix all parameters except one;
- (ii) provide  $d$  independent observables  $F_1, \dots, F_d$ ;
- (iii) impose a normalization, variational principle, or statistical prior;
- (iv) select a discrete parameter by an external integrality condition.

This elementary point is especially important for  $q$ -special functions, where “argument”, “base”, “order”, and “shape parameter” are all plausible inverse variables.

**Definition 1.1** (Partial inverse branch). Fix auxiliary parameters  $\lambda$ , write  $F(p; \lambda)$ , and choose a regular point  $(p_0, \lambda_0)$ . A partial inverse branch is a function  $p = G(y; \lambda)$  satisfying

$$F(G(y; \lambda); \lambda) = y, \quad G(F(p_0; \lambda_0); \lambda_0) = p_0,$$

together with a specified real interval or complex continuation path.

### 1.2 Real branches and complex coverings

If  $F: I \rightarrow \mathbb{R}$  is continuous and strictly monotone, it has one ordinary real inverse on  $F(I)$ . In the complex plane the natural object is instead

$$\mathcal{R}_F = \{(p, y) \in \mathbb{C}^2 : F(p) = y\},$$

projected to the  $y$ -plane. At a regular point  $F_p \neq 0$ , the projection is locally biholomorphic. A point with  $F_p = 0$  creates a branch value; a loop around it permutes inverse sheets. Finite  $q$ -Pochhammer products and Gaussian coefficients give algebraic coverings. Infinite products and basic hypergeometric functions give infinite-sheeted coverings, often with accumulating singular structure near  $|q| = 1$ .

**Principle 1.2** (Branch data are part of the answer). *A numerical value called “the inverse” is incomplete unless it records at least one of: a real interval, a base point and continuation path, an asymptotic condition, a logarithm index, or a root label transported continuously from a regular parameter value.*

### 1.3 Jacobian identifiability

Suppose  $d$  observables are available:

$$\mathbf{F}(\mathbf{p}) = \mathbf{y}, \quad \mathbf{F} = (F_1, \dots, F_d).$$

**Theorem 1.3** (PROVED Local identifiability). *If the Jacobian*

$$J(\mathbf{p}_0) = \left( \frac{\partial F_i}{\partial p_j}(\mathbf{p}_0) \right)_{i,j=1}^d$$

*is nonsingular, then  $\mathbf{p}$  is a locally analytic function of  $\mathbf{y}$ . The first-order sensitivity is*

$$d\mathbf{p} = J(\mathbf{p})^{-1} d\mathbf{y}.$$

*If  $J$  is singular, either identifiability fails or a fractional-power inverse must be used after resolving the singularity.*

*Proof.* This is the analytic inverse-function theorem. Differentiating  $\mathbf{F}(\mathbf{p}(\mathbf{y})) = \mathbf{y}$  gives the sensitivity formula.  $\square$

For noisy data with covariance matrix  $\Sigma_y$ , the linearized covariance is

$$\Sigma_p \approx J^{-1} \Sigma_y (J^{-1})^*.$$

Thus the smallest singular value of a suitably scaled Jacobian measures local recoverability. Near a fold it tends to zero like the square root of the distance to the critical value; near a cusp it vanishes faster.

### 1.4 Dimensionless condition numbers

For a scalar inverse  $p = G(y)$ , define the absolute and relative condition numbers

$$\kappa_{\text{abs}} = |G'(y)| = \frac{1}{|F'(p)|}, \quad \kappa_{\text{rel}} = \left| \frac{y}{pF'(p)} \right|.$$

A large condition number is not a defect in an algorithm; it is an intrinsic property of the inverse problem. The right numerical response is to change coordinates, add an observable, or return a validated set rather than a misleading decimal.

## Chapter 2

# Regular analytic reversion

### 2.1 The scalar inverse jet

Let  $y_0 = f(x_0)$ , put  $\Delta = y - y_0$ , and write

$$f(x_0 + u) = y_0 + f_1 u + \frac{f_2}{2!} u^2 + \frac{f_3}{3!} u^3 + \cdots, \quad f_r = f^{(r)}(x_0).$$

When  $f_1 \neq 0$ , the inverse has a convergent series

$$u = c_1 \Delta + c_2 \Delta^2 + c_3 \Delta^3 + \cdots.$$

**Theorem 2.1** (PROVED Bell-polynomial recursion). *The coefficients are determined by*

$$c_1 = \frac{1}{f_1}, \quad c_n = -\frac{1}{n! f_1} \sum_{r=2}^n f_r B_{n,r}(1!c_1, 2!c_2, \dots, (n-r+1)!c_{n-r+1}),$$

where  $B_{n,r}$  is the exponential partial Bell polynomial. In particular,

$$\begin{aligned} c_2 &= -\frac{f_2}{2f_1^3}, \\ c_3 &= \frac{3f_2^2 - f_1 f_3}{6f_1^5}, \\ c_4 &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{24f_1^7}, \\ c_5 &= \frac{105f_2^4 - 105f_1 f_2^2 f_3 + 10f_1^2 f_3^2 + 15f_1^2 f_2 f_4 - f_1^3 f_5}{120f_1^9}. \end{aligned}$$

*Proof.* Substitute  $u(\Delta) = \sum_{n \geq 1} c_n \Delta^n$  into the Taylor series for  $f$ . The coefficient of  $\Delta^n$  in  $u^r$  is

$$\frac{1}{n!} B_{n,r}(1!c_1, 2!c_2, \dots).$$

The coefficient of  $\Delta$  gives  $c_1$ ; for  $n \geq 2$ , the term  $f_1 c_n$  is the only occurrence of  $c_n$ , so it can be isolated recursively.  $\square$



## 2.2 Lagrange–Bürmann form

If  $y - y_0 = u/\phi(u)$ , with  $\phi(0) \neq 0$ , then

$$u = \sum_{n \geq 1} \frac{(y - y_0)^n}{n!} \left[ \frac{d^{n-1}}{du^{n-1}} \phi(u)^n \right]_{u=0}.$$

More generally, for analytic  $H$ ,

$$H(u(y)) = H(0) + \sum_{n \geq 1} \frac{(y - y_0)^n}{n!} \left[ \frac{d^{n-1}}{du^{n-1}} H'(u) \phi(u)^n \right]_{u=0}.$$

This is often the cleanest route when a  $q$ -product factors into a simple zero times a regular unit.

## 2.3 Logarithmic reversion

Products should normally be inverted after taking a logarithm. If

$$\log \frac{F(p)}{F(p_0)} = A_1 u + A_2 u^2 + A_3 u^3 + \cdots, \quad u = p - p_0,$$

and  $s = \log(y/F(p_0))$ , then the same recursion applies to  $s = A_1 u + A_2 u^2 + \cdots$ . This has three advantages:

- (a) products become sums and derivatives become Lambert series;
- (b) relative error, rather than absolute scale, is controlled;
- (c) reciprocal or palindromic symmetry frequently eliminates all odd or all even terms after centering.

For the common form

$$s = Au + Bu^2 + O(u^4),$$

with no cubic term, one obtains

$$u = \frac{s}{A} - \frac{B}{A^3} s^2 + \frac{2B^2}{A^5} s^3 + O(s^4).$$

## 2.4 Auxiliary-parameter derivatives

Let  $F(G(y, \lambda), \lambda) = y$ . Subscripts denote total derivatives after substitution of  $p = G(y, \lambda)$ .

**Proposition 2.2** (PROVED First and second inverse sensitivities). *At a regular point,*

$$\begin{aligned} G_y &= \frac{1}{F_p}, & G_{\lambda_i} &= -\frac{F_{\lambda_i}}{F_p}, \\ G_{yy} &= -\frac{F_{pp}}{F_p^3}, & G_{y\lambda_i} &= \frac{F_{pp}F_{\lambda_i} - F_p F_{p\lambda_i}}{F_p^3}, \\ G_{\lambda_i \lambda_j} &= -\frac{F_{pp}G_{\lambda_i}G_{\lambda_j} + F_{p\lambda_i}G_{\lambda_j} + F_{p\lambda_j}G_{\lambda_i} + F_{\lambda_i \lambda_j}}{F_p}. \end{aligned}$$

*Proof.* Differentiate the defining identity once and twice. No independent inverse formula is needed; higher jets follow recursively by multivariate Faà di Bruno.  $\square$

## 2.5 Multivariate Lagrange–Good inversion

A convenient canonical system is

$$w_i = z_i \phi_i(\mathbf{w}), \quad i = 1, \dots, d,$$

with  $\phi_i(\mathbf{0}) \neq 0$ . Good's formula [7] states that for a formal power series  $H$ ,

$$H(\mathbf{w}(\mathbf{z})) = \sum_{\mathbf{n} \in \mathbb{N}^d} \mathbf{z}^{\mathbf{n}} [\mathbf{x}^{\mathbf{n}}] H(\mathbf{x}) \prod_{i=1}^d \phi_i(\mathbf{x})^{n_i} \det \left( \delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right).$$

A general locally invertible vector equation can be brought to this form after a linear change of variables. In practice, however, a coefficient-by-coefficient Jacobian recursion is often simpler to implement and easier to certify.

## Chapter 3

# Singular inversion and asymptotic reversion

### 3.1 Simple folds and Puiseux coefficients

Assume

$$f'(x_*) = 0, \quad f''(x_*) = f_2 \neq 0, \quad y_* = f(x_*).$$

Then the inverse is not analytic in  $y - y_*$ , but it is analytic in a square root.

**Theorem 3.1** (PROVED Fold normal form). *With  $\Delta = y - y_*$ , the two local sheets are*

$$\begin{aligned} x - x_* = & a_1 \Delta^{1/2} - \frac{f_3}{3f_2^2} \Delta + a_1 \frac{5f_3^2 - 3f_2 f_4}{36f_2^3} \Delta^{3/2} \\ & + \frac{-9f_2^2 f_5 + 45f_2 f_3 f_4 - 40f_3^3}{270f_2^5} \Delta^2 + O(\Delta^{5/2}), \end{aligned}$$

where  $a_1 = \pm \sqrt{2/f_2}$ , with compatible branches of the square root.

*Proof.* Put  $x - x_* = a_1 s + a_2 s^2 + a_3 s^3 + \dots$ ,  $s^2 = \Delta$ , substitute into the Taylor series, and solve successively. The coefficient of  $s^2$  gives  $a_1^2 f_2 / 2 = 1$ ; the remaining displayed coefficients follow by triangular elimination.  $\square$

More generally, if

$$f'(x_*) = \dots = f^{(r-1)}(x_*) = 0, \quad f^{(r)}(x_*) \neq 0,$$

then  $r$  sheets meet and

$$x - x_* \sim \left( \frac{r!(y - y_*)}{f^{(r)}(x_*)} \right)^{1/r}.$$

Continuation once around  $y_*$  cycles these sheets.

### 3.2 Newton polygons for mixed powers

Near  $q = 0$ , continuous parameters often produce generalized expansions such as

$$s = c_1 q^\alpha + c_2 q + c_3 q^{2\alpha} + \dots, \quad 0 < \alpha < 1.$$

The next correction is not determined by the visual order in which the terms were written; it is determined by the lower convex hull of exponent pairs. The resulting inverse is a Hahn or Puiseux series whose exponents lie in the additive semigroup generated by 1 and  $\alpha$ . Rational  $\alpha$  produces an ordinary Puiseux series after a finite ramification. Irrational  $\alpha$  produces a well-ordered generalized series but no finite ramification.

### 3.3 Large-log inversion

A recurring endpoint form is

$$\log y = -\frac{A}{t} + C - Dt + Et^3 - Ft^5 + \dots, \quad A \neq 0,$$

with  $t \rightarrow 0$ . Put  $X = -\log y + C$ . Formal reversion gives

$$t = \frac{A}{X} + \frac{DA^2}{X^3} + \frac{A^3(2D^2 - AE)}{X^5} + \frac{A^4(A^2F - 6ADE + 5D^3)}{X^7} + \dots. \quad (3.1)$$

This formula is the main engine for radial inverses of infinite  $q$ -products.

### 3.4 Logarithmic corrections and Lambert $W$

A second canonical form is

$$\Lambda = \frac{A}{t} + B \log t + C + O(t).$$

Ignoring  $O(t)$  but retaining the logarithm already gives a much better preconditioner than  $A/\Lambda$ . If  $B \neq 0$ , set

$$R = \frac{\Lambda - C}{B} - \log \frac{A}{B}.$$

Then the truncated equation is solved exactly by

$$t = -\frac{A}{B W_k(-e^{-R})}, \quad (3.2)$$

where the branch  $k$  is selected to make  $t$  approach the prescribed ray. This does not replace the full asymptotic series; it supplies an excellent starting value for Newton or interval Newton iteration.

### 3.5 Exponentially small and periodic corrections

Euler–Maclaurin expansions may miss terms of size  $e^{-c/t}$ , while splitting a geometric product at its largest term can produce a bounded periodic function of  $\log x / \log(1/q)$ . Both effects survive inversion. A complete inverse asymptotic may therefore have the transseries form

$$p(y) \sim p_{\text{alg}}(y) + \Omega(\log y) + \sum_{r \geq 1} e^{-rS(p_{\text{alg}})} p_r(y),$$

not a single power series. The archive's numerical algorithms preserve these corrections by evaluating the original product rather than only its formal jet.

## Chapter 4

# A universal theory for positive and palindromic $q$ -polynomials

### 4.1 Positive-base inversion

Let

$$P(q) = \sum_{j=0}^D c_j q^j, \quad c_j \geq 0,$$

with at least one  $c_j > 0$  for  $j > 0$ .

**Theorem 4.1** (**PROVED** Global positive inverse). *The restriction  $P : (0, \infty) \rightarrow (c_0, \infty)$  is strictly increasing and therefore has exactly one inverse for every  $y > c_0$ .*

*Proof.* For  $q > 0$ ,

$$P'(q) = \sum_{j=1}^D j c_j q^{j-1} > 0.$$

Also  $P(q) \rightarrow c_0$  as  $q \downarrow 0$  and  $P(q) \rightarrow \infty$  as  $q \rightarrow \infty$ . □

If  $c_1 > 0$ , the small-base inverse is

$$q = \frac{y - c_0}{c_1} - \frac{c_2}{c_1^3}(y - c_0)^2 + \frac{2c_2^2 - c_1 c_3}{c_1^5}(y - c_0)^3 + O((y - c_0)^4).$$

If  $c_1 = \dots = c_{r-1} = 0$ ,  $c_r > 0$ , the inverse begins with the  $r$ -th root  $((y - c_0)/c_r)^{1/r}$ .

### 4.2 Cumulant coordinates at $q = 1$

Assume  $P(1) > 0$ , and define a random variable  $J$  by

$$\Pr(J = j) = \frac{c_j}{P(1)}.$$

Then

$$\frac{P(e^t)}{P(1)} = \mathbb{E}(e^{tJ}), \quad \log \frac{P(e^t)}{P(1)} = \sum_{r \geq 1} \frac{\kappa_r(J)}{r!} t^r.$$

Thus inversion of  $P$  at  $q = 1$  is exactly cumulant reversion. This probabilistic language is not metaphorical: the coefficients of a Gaussian polynomial enumerate partition areas, and those areas are the random variable  $J$ .

**Theorem 4.2** (PROVED Palindromic reduction). *If  $c_j = c_{D-j}$ , then*

$$\kappa_1(J) = D/2, \quad \kappa_{2r+1}(J) = 0 \quad (r \geq 1).$$

Consequently, with  $s = \log(y/P(1))$ ,

$$s = \frac{D}{2}t + \frac{\kappa_2}{2}t^2 + \frac{\kappa_4}{24}t^4 + \dots,$$

and the positive local inverse is obtained by ordinary reversion in  $t = \log q$ .

*Proof.* Palindromicity makes  $J - D/2$  symmetric in distribution. Its odd moments and odd cumulants vanish.  $\square$

### 4.3 Reciprocity and the inverse at infinity

If

$$P(q) = q^D P(q^{-1}), \quad P(q) = 1 + c_1 q + c_2 q^2 + \dots,$$

set  $R = y^{-1/D}$  on a chosen branch. Reversion at  $Q = q^{-1} = 0$  yields

$$q = R^{-1} - \frac{c_1}{D} + \left( \frac{(D-1)c_1^2}{2D^2} - \frac{c_2}{D} \right) R + O(R^2). \quad (4.1)$$

The formula simultaneously describes  $q \rightarrow +\infty$ ,  $q \rightarrow -\infty$ , and complex rays; the branch of  $R = y^{-1/D}$  labels the sheet.

### 4.4 Cyclotomic zeros

For a polynomial product, a root of unity may force factors to vanish. If

$$P(q) = \Phi_m(q)^\nu U(q), \quad U(\zeta) \neq 0,$$

near a primitive  $m$ -th root  $\zeta$ , then

$$P(q) = (\Phi'_m(\zeta))^\nu U(\zeta) (q - \zeta)^\nu (1 + O(q - \zeta)).$$

Hence the inverse at target zero has  $\nu$  Puiseux sheets,

$$q - \zeta \sim \left( \frac{y}{(\Phi'_m(\zeta))^\nu U(\zeta)} \right)^{1/\nu}.$$

This simple observation controls all root-of-unity inverses of  $q$ -factorials and many of those of Gaussian coefficients.

## **Part II**

# **Finite and infinite $q$ -Pochhammer inverses**

## Chapter 5

# Finite $q$ -Pochhammer inversion in the argument

### 5.1 The polynomial covering in $a$

Fix  $q \neq 0$  and write

$$F_n(a; q) = (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j).$$

If the numbers  $q^{-j}$  are distinct,  $F_n$  is a degree- $n$  polynomial with simple zeros

$$a_j = q^{-j}, \quad 0 \leq j < n.$$

The inverse equation  $F_n(a; q) = y$  therefore has  $n$  complex sheets. Their branch points are the critical values  $F_n(a_*; q)$  with

$$\frac{\partial}{\partial a} \log F_n(a; q) = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j} = 0.$$

The discriminant

$$\text{Disc}_a(F_n(a; q) - y)$$

is a polynomial in  $q$  and  $y$ ; its zero set is the branch curve. For large  $n$ , it is usually better to track critical points numerically than to expand this discriminant.

### 5.2 The inverse jet at $a = 0$

The finite  $q$ -binomial theorem gives

$$(a; q)_n = \sum_{r=0}^n (-a)^r q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q. \quad (5.1)$$

Let  $u = 1 - y$ , and assume  $[n]_q \neq 0$ . Reversion of

$$u = [n]_q a - q \begin{bmatrix} n \\ 2 \end{bmatrix}_q a^2 + q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q a^3 - \dots$$



gives the explicit connection between inverse Pochhammer coefficients and Gaussian coefficients:

$$a = \frac{u}{[n]_q} + \frac{q \begin{bmatrix} n \\ 2 \end{bmatrix}_q}{[n]_q^3} u^2 + \left( \frac{2q^2 \begin{bmatrix} n \\ 2 \end{bmatrix}_q^2}{[n]_q^5} - \frac{q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q}{[n]_q^4} \right) u^3 + O(u^4). \quad (5.2)$$

All higher terms are obtained from the Bell recursion in [Theorem 5.1](#), below.

**Theorem 5.1** ([PROVED](#) All-order argument inverse at one). *Let*

$$b_r = (-1)^{r+1} q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q, \quad u = \sum_{r \geq 1} b_r a^r.$$

If  $b_1 = [n]_q \neq 0$ , the coefficient of  $u^N$  in the inverse  $a(u)$  is

$$[u^N]a(u) = \frac{1}{N} [z^{N-1}] \left( \frac{z}{\sum_{r \geq 1} b_r z^r} \right)^N.$$

*Proof.* This is Lagrange–Bürmann applied to  $u = a/\phi(a)$ , with  $\phi(a) = a/u(a)$ . □

At a cyclotomic base with  $[n]_q = 0$ , the branch at  $a = 0$  is singular. If the first nonzero coefficient in (5.1) has index  $r$ , then  $a \asymp (1 - y)^{1/r}$ . Thus the same root-of-unity geometry appears whether one inverts in  $a$  or in  $q$ .

### 5.3 Local inverse at a simple zero

Assume  $q^r \neq q^s$  for  $0 \leq r < s < n$ . At  $a_r = q^{-r}$ ,

$$F'_n(a_r; q) = (-1)^{r+1} q^{-r(r-1)/2} (q; q)_r (q; q)_{n-1-r}. \quad (5.3)$$

**Proposition 5.2** ([PROVED](#) Zero-sheet expansion). *For each  $r$ , the inverse branch satisfying  $a(0) = q^{-r}$  is*

$$a(y) = q^{-r} + \frac{y}{F'_n(q^{-r}; q)} - \frac{F''_n(q^{-r}; q)}{2F'_n(q^{-r}; q)^3} y^2 + O(y^3).$$

*Proof.* The generic inverse coefficients apply because (5.3) is nonzero. To derive that derivative, remove the vanishing factor  $1 - aq^r$ . For  $j < r$ , use  $1 - q^{j-r} = -q^{j-r}(1 - q^{r-j})$ , and multiply the resulting powers of  $q$ . □

### 5.4 Colliding zeros at roots of unity

Let  $\zeta$  be primitive of order  $m$ . At  $q = \zeta$ , the zero  $a = \zeta^{-r}$  is repeated once for every index  $j \in \{0, \dots, n-1\}$  congruent to  $r \pmod{m}$ . Its multiplicity is

$$\nu_r(n, m) = \begin{cases} 1 + \lfloor (n-1-r)/m \rfloor, & 0 \leq r < \min(m, n), \\ 0, & r \geq n. \end{cases}$$

**Theorem 5.3** (PROVED Cyclotomic argument ramification). *At  $q = \zeta$ , write*

$$(a; \zeta)_n = (1 - a\zeta^r)^{\nu_r} U_r(a), \quad U_r(\zeta^{-r}) \neq 0.$$

*Then the inverse branches at target  $y = 0$  satisfy*

$$a - \zeta^{-r} \sim -\zeta^{-r} \left( \frac{y}{U_r(\zeta^{-r})} \right)^{1/\nu_r}.$$

*Thus the ramification index is exactly the occupancy of one congruence class of exponents modulo  $m$ .*

*Proof.* All factors with  $j \equiv r \pmod{m}$  coincide with  $1 - a\zeta^r$ ; all others are nonzero at  $a = \zeta^{-r}$ . Apply the multiplicity form of the Puiseux theorem.  $\square$

## 5.5 Exact inverses at $q = 1$ and $q = -1$

At  $q = 1$ ,

$$(a; 1)_n = (1 - a)^n, \quad a = 1 - \omega_n^j y^{1/n}, \quad 0 \leq j < n.$$

At  $q = -1$ , put

$$E = \lceil n/2 \rceil, \quad O = \lfloor n/2 \rfloor.$$

Then

$$(a; -1)_n = (1 - a)^E (1 + a)^O. \tag{5.4}$$

For even  $n = 2m$ , this is  $(1 - a^2)^m$ , so every inverse sheet is explicit:

$$a = \pm \sqrt{1 - \omega_m^j y^{1/m}}, \quad 0 \leq j < m.$$

At  $y = 1$ , all sheets pair at  $a = 0$ , and the real inverse has the expected square-root singularity.

For odd  $n = 2m + 1$ , the derivative of  $(1 - a)^{m+1}(1 + a)^m$  vanishes at

$$a = -\frac{1}{2m+1}.$$

The corresponding critical value is

$$Y_m = \left( \frac{2m+2}{2m+1} \right)^{m+1} \left( \frac{2m}{2m+1} \right)^m. \tag{5.5}$$

The two real argument branches merge there in a quadratic fold. This same parameter will reappear as the critical *base* parameter at  $q = -1$ , a nonaccidental intersection studied in [Section 6.4](#).

## 5.6 Large-base clustering of argument sheets

For fixed  $n$ , the zeros are  $1, q^{-1}, \dots, q^{1-n}$ . As  $|q| \rightarrow \infty$ , one zero remains near  $a = 1$ , while the other  $n - 1$  zeros collapse toward the origin on distinct scales. The expansion (5.1) shows that a branch with bounded target and  $a \rightarrow 0$  typically has

$$a \sim \frac{1-y}{[n]_q} \sim (1-y)q^{1-n}.$$

Branches attached to the zero  $q^{-r}$  should instead be rescaled by  $a = q^{-r}(1 + u)$ . The resulting Newton polygon separates the root cluster into  $n - 1$  geometric layers. This is a useful example where a single unscaled expansion in  $a$  cannot be uniform across all inverse sheets.

## Chapter 6

# Finite $q$ -Pochhammer inversion in the base

### 6.1 Logarithmic derivatives and the critical set

For fixed  $a$ ,

$$\partial_q \log(a; q)_n = -a \sum_{j=1}^{n-1} \frac{jq^{j-1}}{1 - aq^j}, \quad (6.1)$$

$$\partial_a^r \log(a; q)_n = -(r-1)! \sum_{j=0}^{n-1} \frac{q^{jr}}{(1 - aq^j)^r}. \quad (6.2)$$

A base inverse branch is regular precisely when the first expression is finite and nonzero. Multiplying it by the product denominator produces an explicit polynomial equation for the critical set in  $(a, q)$ . Eliminating  $q$  from that equation and  $(a; q)_n - y = 0$  gives the branch-value discriminant.

An alternative expansion, valid analytically where the sums converge and formally elsewhere, is

$$\log(a; q)_n = - \sum_{r \geq 1} \frac{a^r}{r} \frac{1 - q^{nr}}{1 - q^r}. \quad (6.3)$$

It is often the most efficient source of high derivatives with respect to  $q$ .

### 6.2 All-order inversion at $q = 0$

Assume  $n \geq 2$  and  $a \notin \{0, 1\}$ . Factor the constant term:

$$\frac{(a; q)_n}{1 - a} = \prod_{j=1}^{n-1} (1 - aq^j) = \exp \left( \sum_{r \geq 1} \ell_r^{(n)}(a) q^r \right),$$

where

$$\ell_r^{(n)}(a) = - \sum_{\substack{d|r \\ 1 \leq d \leq n-1}} \frac{a^{r/d}}{r/d}. \quad (6.4)$$

The forward coefficients are complete Bell polynomials in the  $\ell_r$ , and the inverse coefficients follow from the scalar recursion. This is an all-order arithmetic description: divisors of the exponent control each jet.

For  $n \geq 4$ , define

$$\eta = \frac{(1 - a) - y}{a(1 - a)}.$$

Then

$$q = \eta - \eta^2 + (1 + a)\eta^3 + O(\eta^4). \quad (6.5)$$

The cubic coefficient changes to  $2 + a$  for  $n = 3$ , while for  $n = 2$  the relation is exactly  $q = \eta$ . For the infinite product, the stable all-order coefficients begin

$$q = \eta - \eta^2 + (1 + a)\eta^3 - (1 + 4a)\eta^4 + (1 + 11a + 3a^2)\eta^5 + O(\eta^6). \quad (6.6)$$

### 6.3 All-order inversion at $q = 1$

Use the exponentially natural coordinate  $q = e^{-t}$ . Put

$$S_r(n) = \sum_{j=0}^{n-1} j^r.$$

From

$$\log(1 - ae^{-x}) = \log(1 - a) + \sum_{r \geq 1} \frac{(-1)^{r+1}}{r!} \text{Li}_{1-r}(a)x^r,$$

we obtain the exact formal jet

$$\log \frac{(a; e^{-t})_n}{(1 - a)^n} = \sum_{r \geq 1} A_r t^r, \quad A_r = \frac{(-1)^{r+1}}{r!} \text{Li}_{1-r}(a) S_r(n). \quad (6.7)$$

When  $A_1 \neq 0$ , let

$$s = \log \frac{y}{(1 - a)^n}.$$

Then

$$t = \frac{s}{A_1} - \frac{A_2}{A_1^3} s^2 + \frac{2A_2^2 - A_1 A_3}{A_1^5} s^3 + O(s^4), \quad q = e^{-t}. \quad (6.8)$$

This formula applies to complex  $a$ , provided the logarithms and polylogarithms are continued consistently. The degeneracies  $a = 0$  and  $n = 1$  correspond to a product independent of  $q$ . The value  $a = 1$  puts the target at an identically vanishing factor and requires a joint scaling of  $a - 1$  and  $q - 1$ .

### 6.4 The complete $q = -1$ critical classification

At  $q = -1$ , recall

$$F_n(a, -1) = (1 - a)^E (1 + a)^O, \quad E = \lceil n/2 \rceil, \quad O = \lfloor n/2 \rfloor.$$

Evaluating (6.1) gives

$$\partial_q \log F_n(a, q)|_{q=-1} = \frac{aE(E-1)}{1-a} - \frac{aO^2}{1+a}. \quad (6.9)$$

**Theorem 6.1** (PROVED Exact nonconstant critical parameter at  $q = -1$ ). *Let  $n \geq 3$ , and exclude  $a = 0, \pm 1$  when they make the slice constant or singular. The fixed- $a$  function  $q \mapsto (a; q)_n$  is critical at  $q = -1$  if and only if*

$$a = a_*(n) = \begin{cases} \frac{1}{n-1}, & n \text{ even}, \\ -\frac{1}{n}, & n \text{ odd}. \end{cases}$$

For even  $n = 2m$ ,  $m \geq 2$ ,

$$F_* = \left( \frac{4m(m-1)}{(2m-1)^2} \right)^m, \quad (6.10)$$

$$\partial_q^2 \log F_n|_* = -\frac{(m+1)(2m-1)^2}{12m(m-1)} \neq 0. \quad (6.11)$$

For odd  $n = 2m+1$ ,  $m \geq 2$ , the critical value is  $Y_m$  from (5.5), and

$$\partial_q^2 \log F_n|_* = -\frac{(m-1)(2m+1)^2}{12m(m+1)} \neq 0. \quad (6.12)$$

Thus all these inverse branches are quadratic folds. The sole nonconstant higher degeneracy is

$$(a; q)_3|_{a=-1/3} = \frac{32}{27} + \frac{4}{27}(q+1)^3. \quad (6.13)$$

*Proof.* Equation (6.9) is zero, apart from the constant case  $a = 0$ , precisely when

$$E(E-1)(1+a) = O^2(1-a).$$

For  $n = 2m$ , this gives  $a = 1/(2m-1)$ ; for  $n = 2m+1$ , it gives  $a = -1/(2m+1)$ . Substitution into  $(1-a)^E(1+a)^O$  gives the critical values. Differentiating (6.1) once more and summing the even and odd arithmetic progressions gives (6.11) and (6.12). The odd formula vanishes only at  $m = 1$ . Direct factorization at  $n = 3$  gives (6.13). No other solution exists because the critical equation is linear in  $a$  after removing the constant factor  $a$ .  $\square$

The fold inverse follows immediately. Since  $F_q = 0$  at the critical point,  $F_{qq} = F_* \partial_q^2 \log F$ , and therefore

$$q+1 = \pm \sqrt{\frac{2(y-F_*)}{F_{qq}(a_*, -1)}} + O(y-F_*).$$

For  $n = 3$ , the inverse is exact:

$$q = -1 + \left( \frac{27}{4} \left( y - \frac{32}{27} \right) \right)^{1/3}.$$

## 6.5 The exceptional cubic is a Morse saddle in two parameters

For odd  $n = 2m+1$ , the same point

$$(a, q) = \left( -\frac{1}{2m+1}, -1 \right)$$

also satisfies  $F_a = 0$ , as follows from (5.4). At a point where both first derivatives vanish, the Hessian of  $F$  is  $F_*$  times the Hessian of  $\log F$ . Its entries are

$$\begin{aligned} L_{aa} &= -\frac{(2m+1)^3}{4m(m+1)}, \\ L_{aq} &= -\frac{(2m+1)^2}{4(m+1)}, \\ L_{qq} &= -\frac{(m-1)(2m+1)^2}{12m(m+1)}, \end{aligned}$$

and

$$\det \nabla^2 L = -\frac{(2m+1)^4(m^2+m+1)}{48m^2(m+1)^2} < 0. \quad (6.14)$$

**Corollary 6.2** (PROVED Morse interpretation). *Every odd critical point above, including  $n = 3$ , is a nondegenerate saddle of the two-variable map  $(a, q) \mapsto (a; q)_n$ . The cubic inverse in (6.13) is a nongeneric one-dimensional slice tangent to a null direction of the indefinite quadratic form; it is not a cusp of the full parameter surface.*

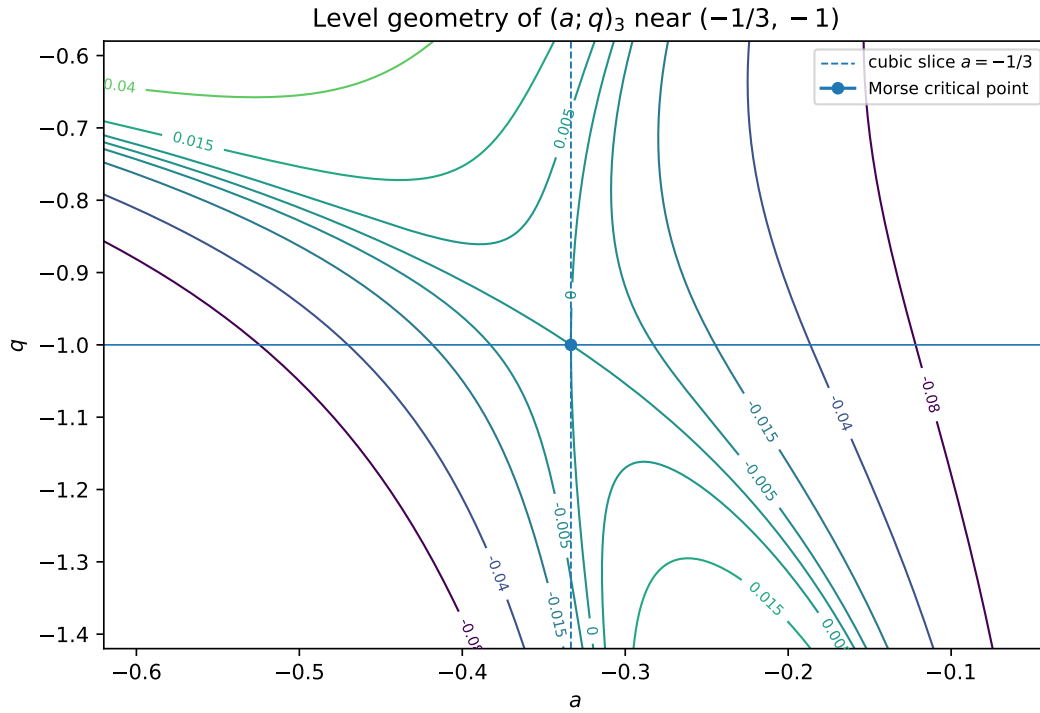


Figure 6.1: Level geometry of  $(a; q)_3$  near  $(-1/3, -1)$ . The vertical fixed- $a$  slice begins cubically, while the full two-parameter Hessian is indefinite and nonsingular.

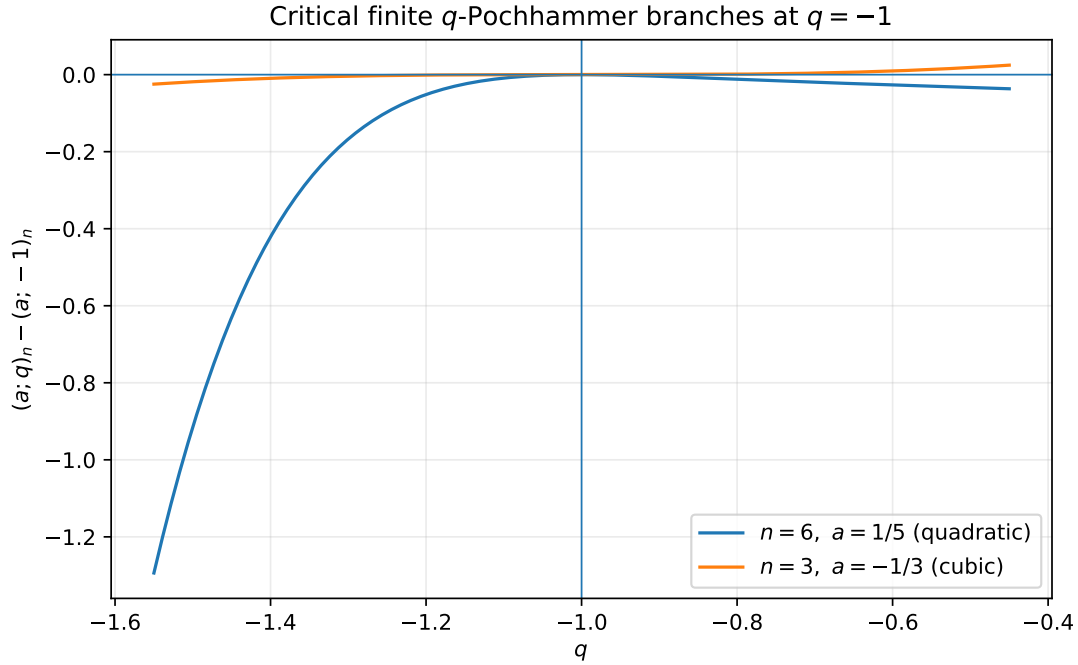


Figure 6.2: A quadratic critical slice at  $n = 6, a = 1/5$  and the exceptional cubic slice at  $n = 3, a = -1/3$ , both centered by subtracting their value at  $q = -1$ .

## 6.6 Other roots of unity

Let  $\zeta$  be primitive of order  $m$ , and write  $n = mN + r$ ,  $0 \leq r < m$ . Periodicity gives the exact block formula

$$(a; \zeta)_n = (1 - a^m)^N (a; \zeta)_r. \quad (6.15)$$

A regular base inverse has

$$q = \zeta + \frac{y - (a; \zeta)_n}{\partial_q (a; q)_n|_{q=\zeta}} + O((y - y_0)^2).$$

The critical locus is the rational equation

$$\sum_{j=1}^{n-1} \frac{ja\zeta^{j-1}}{1 - a\zeta^j} = 0,$$

after cancellation of any factors vanishing at  $a^m = 1$ . Multiplication by  $\prod_j (1 - a\zeta^j)$  shows that it is algebraic over the cyclotomic field  $\mathbb{Q}(\zeta)$ .

**Research problem 6.3.** Classify the critical parameters  $a$  and their degeneracy at every primitive root of unity. The  $m = 2$  theorem above suggests that the generic critical points are folds and that higher degeneracies arise only from special low-dimensional tangencies or colliding product zeros.

## 6.7 The inverse at $q = \pm\infty$

Let  $N = \binom{n}{2}$ , assume  $n \geq 3$ , and put

$$C = (1 - a)(-a)^{n-1}, \quad Q = q^{-1}.$$

Then

$$(a; q)_n = Cq^N \prod_{j=1}^{n-1} (1 - a^{-1}Q^j) = Cq^N (1 - a^{-1}Q - a^{-1}Q^2 + O(Q^3)).$$

For a selected branch

$$R = \left( \frac{C}{y} \right)^{1/N},$$

reversion gives

$$q = R^{-1} + \frac{a^{-1}}{N} + O(R). \tag{6.16}$$

Further terms follow from the divisor/Bell machinery at  $Q = 0$ . Real  $q \rightarrow -\infty$  is possible only on target sign sectors compatible with  $C(-1)^N$ ; in the complex plane all  $N$  branches are present.



## Chapter 7

# Order inversion and simultaneous recovery for finite products

### 7.1 Discrete order

For integer order,

$$(a; q)_{n+1} = (a; q)_n (1 - aq^n).$$

A discrete inverse in  $n$  is therefore a threshold or search problem, not an analytic inverse, unless an interpolation is chosen. On a real sign domain where all factors lie in  $(0, 1)$ , the sequence decreases strictly and the minimal  $n$  satisfying  $(a; q)_n \leq y$  is unambiguous. Outside such a domain, oscillation and repeated values require a branch label even at the discrete level.

### 7.2 Complex order and exact reduction to argument inversion

For  $|q| < 1$ , define

$$(a; q)_\alpha = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty},$$

with a chosen branch of  $q^\alpha$ . Let  $\mathcal{A}_q(z)$  denote a selected argument inverse of  $(b; q)_\infty = z$ . Then

**Theorem 7.1** (PROVED Exact order inverse). *Every order branch satisfying  $(a; q)_\alpha = y$  is of the form*

$$\alpha = \frac{\text{Log}(\mathcal{A}_q((a; q)_\infty/y)/a) + 2\pi i k}{\text{Log } q}, \quad k \in \mathbb{Z}, \quad (7.1)$$

with compatible logarithm and  $\mathcal{A}_q$  sheets.

*Proof.* The equation is equivalent to  $(aq^\alpha; q)_\infty = (a; q)_\infty/y$ . Invert in the argument, then take a logarithm of  $q^\alpha = \mathcal{A}_q(\cdot)/a$ .  $\square$

The order derivative is the Lambert series

$$\partial_\alpha \log(a; q)_\alpha = (\log q) \sum_{j \geq 0} \frac{aq^{\alpha+j}}{1 - aq^{\alpha+j}}. \quad (7.2)$$

It supplies local inverse jets and quantifies ill-conditioning when the argument inverse approaches a critical point.

### 7.3 Exact recovery from consecutive products

Suppose  $P_n = (a; q)_n$ ,  $P_{n+1}$ , and  $P_{n+2}$  are observed exactly, with no zero denominator. Define

$$r_n = 1 - \frac{P_{n+1}}{P_n}, \quad r_{n+1} = 1 - \frac{P_{n+2}}{P_{n+1}}.$$

Then  $r_n = aq^n$  and  $r_{n+1} = aq^{n+1}$ .

**Theorem 7.2** (**PROVED** Two-parameter reconstruction). *If  $r_n \neq 0$ ,*

$$q = \frac{r_{n+1}}{r_n}, \quad a = r_n q^{-n}. \quad (7.3)$$

This exact formula is preferable to solving one high-degree equation. With noise, the ratios should be fitted jointly across many consecutive orders; the logarithms

$$\log r_j = \log a + j \log q$$

turn the problem into a linear regression after branch unwrapping.

### 7.4 General two-observable inversion

If only two nonconsecutive measurements

$$F_{n_1}(a, q) = y_1, \quad F_{n_2}(a, q) = y_2$$

are available, local identifiability is controlled by

$$J = \det \begin{pmatrix} \partial_a F_{n_1} & \partial_q F_{n_1} \\ \partial_a F_{n_2} & \partial_q F_{n_2} \end{pmatrix}.$$

The locus  $J = 0$  is an experimental-design discriminant: measurements there cannot separate changes in  $a$  from changes in  $q$  to first order. One should choose orders whose logarithmic derivative vectors are as far from parallel as possible.

## Chapter 8

# The infinite $q$ -Pochhammer inverse in its argument

### 8.1 Principal real branch

Fix  $0 < q < 1$ . On  $a < 1$ ,

$$P_q(a) = (a; q)_\infty > 0, \quad \partial_a \log P_q(a) = - \sum_{j \geq 0} \frac{q^j}{1 - aq^j} < 0.$$

Moreover,  $P_q(a) \rightarrow 0$  as  $a \uparrow 1$ , while  $P_q(a) \rightarrow \infty$  as  $a \rightarrow -\infty$ . Hence there is a unique decreasing bijection

$$P_q : (-\infty, 1) \longrightarrow (0, \infty).$$

We denote its inverse by

$$a = \mathcal{A}_q(y).$$

Other real branches lie between successive positive zeros  $q^{-j}$ , and the complex inverse has infinitely many sheets.

### 8.2 Near target one: argument close to zero

The logarithmic series

$$\log(a; q)_\infty = - \sum_{r \geq 1} \frac{a^r}{r(1 - q^r)}$$

gives

$$(a; q)_\infty = 1 - \frac{a}{1 - q} + \frac{q}{(1 - q)^2(1 + q)} a^2 + O(a^3).$$

With  $u = 1 - y$ , the principal inverse is

$$\mathcal{A}_q(y) = (1 - q)u + \frac{q(1 - q)}{1 + q} u^2 + O(u^3). \tag{8.1}$$

The full series is obtained by Lagrange inversion of the Lambert exponential. Its radius of convergence is the distance in the  $y$ -plane to the nearest critical value of the entire function  $a \mapsto (a; q)_\infty$ .

### 8.3 Near target zero: argument close to one

Write  $a = 1 - \varepsilon$ . Then

$$(a; q)_\infty = \varepsilon(aq; q)_\infty.$$

Define Lambert sums

$$L_1(q) = \sum_{j \geq 1} \frac{q^j}{1 - q^j}, \quad L_2(q) = \sum_{j \geq 1} \frac{q^{2j}}{(1 - q^j)^2}.$$

Taylor expansion at  $a = 1$  gives

$$(1 - \varepsilon; q)_\infty = (q; q)_\infty \varepsilon \left( 1 + L_1(q)\varepsilon + \frac{L_1(q)^2 - L_2(q)}{2} \varepsilon^2 + \dots \right).$$

Therefore, with  $v = y/(q; q)_\infty$ ,

$$\mathcal{A}_q(y) = 1 - v + L_1(q)v^2 + O(v^3). \quad (8.2)$$

This expansion is especially effective for inverse Euler  $q$ -exponentials.

### 8.4 Large positive target and the negative argument sheet

Set  $a = -x$ ,  $x > 0$ , and  $L = \log(1/q)$ . Splitting the product near the index  $j \approx \log x/L$ , or using a theta-function completion, yields an asymptotic of the form

$$\log(-x; q)_\infty = \frac{(\log x)^2}{2L} + \frac{1}{2} \log x + C_q + \Omega_q \left( \frac{\log x}{L} \right) + O(x^{-1}), \quad (8.3)$$

where  $\Omega_q$  is bounded and one-periodic. Hence

$$\log x = \sqrt{2L \log y} - \frac{L}{2} + O(1), \quad \mathcal{A}_q(y) = -\exp \left( \sqrt{2L \log y} - \frac{L}{2} + O(1) \right).$$

The bounded  $O(1)$  term contains a log-periodic correction; dropping it can produce a relative error that does not tend monotonically to zero. The prior inverse report develops this branch in greater numerical detail [3]; here it serves as a prototype of periodic inverse asymptotics.

### 8.5 Nonprincipal real branch points

Between consecutive zeros  $q^{-j}$  and  $q^{-j-1}$ , the real product has at least one critical point. For large  $j$ , scale

$$a = q^{-j}b.$$

The local product separates into a finite sign-changing block and a stable tail, producing an asymptotic lattice of branch values. A systematic description uses the same splitting that produces  $\Omega_q$ . The branch index becomes a discrete phase variable coupled to the continuous target. This is an instance of an inverse covering with asymptotically self-similar sheets.

## Chapter 9

# The infinite product inverted in the base

### 9.1 Near $q = 0$

For the infinite product, all coefficients in (6.6) are stable because only finitely many factors contribute to each power of  $q$ . Explicitly, with

$$\eta = \frac{(1-a) - y}{a(1-a)},$$

$$q = \eta - \eta^2 + (1+a)\eta^3 - (1+4a)\eta^4 + (1+11a+3a^2)\eta^5 - (1+26a+22a^2)\eta^6 + O(\eta^7). \quad (9.1)$$

The divisor formula (6.4), now with no upper restriction on  $d$ , and Bell reversion generate arbitrary order. Singular cases occur at  $a = 0$  and  $a = 1$ , where the normalization by  $a(1-a)$  fails because the forward map degenerates.

### 9.2 Fixed argument and $q \rightarrow 1^-$

Let  $q = e^{-t}$ ,  $t \downarrow 0$ , and keep  $|a| < 1$  fixed. Expanding  $1/(1 - e^{-rt})$  in the Lambert series gives the classical McIntosh-type expansion [10, 11]

$$\log(a; e^{-t})_\infty = -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1-a) - \sum_{r \geq 1} \frac{B_{2r}}{(2r)!} \text{Li}_{2-2r}(a) t^{2r-1}. \quad (9.2)$$

Thus

$$A = \text{Li}_2(a), \quad D = \frac{\text{Li}_0(a)}{12}, \quad E = \frac{\text{Li}_{-2}(a)}{720}, \quad F = \frac{\text{Li}_{-4}(a)}{30240}$$

in (3.1). With

$$X = -\log y + \frac{1}{2} \log(1-a),$$

the base inverse is

$$t = \frac{A}{X} + \frac{DA^2}{X^3} + \frac{A^3(2D^2 - AE)}{X^5} + \frac{A^4(A^2F - 6ADE + 5D^3)}{X^7} + O(X^{-9}), \quad q = e^{-t}. \quad (9.3)$$

For real  $0 < a < 1$ ,  $y \downarrow 0$  and all displayed quantities are real. For complex  $a$ , this formula labels branches through the choices of  $\text{Li}_2$ ,  $\log(1-a)$ , and  $\log y$ .

### 9.3 The nonuniform scaling $a = q^x$

When  $a \rightarrow 1$  together with  $q \rightarrow 1$ , the fixed- $a$  expansion is not uniform. Put  $a = q^x = e^{-xt}$ , with fixed  $x \notin \{0, -1, -2, \dots\}$ . A Bernoulli-polynomial expansion is

$$\begin{aligned} \log(q^x; q)_\infty &= -\frac{\pi^2}{6t} + \left(\frac{1}{2} - x\right) \log t + \frac{1}{2} \log(2\pi) - \log \Gamma(x) \\ &\quad - \sum_{r \geq 1} \frac{B_r B_{r+1}(x)}{r(r+1)!} t^r. \end{aligned} \quad (9.4)$$

The first correction is  $B_2(x)t/4$ .

Let  $\Lambda = -\log y$ ,  $A = \pi^2/6$ ,

$$B = x - \frac{1}{2}, \quad C = -\frac{1}{2} \log(2\pi) + \log \Gamma(x).$$

Then

$$\Lambda = \frac{A}{t} + B \log t + C - \frac{B_2(x)}{4} t + \dots$$

A first explicit approximation is

$$t \sim \frac{A}{\Lambda + B \log \Lambda - B \log A - C}. \quad (9.5)$$

Retaining the logarithm exactly gives the Lambert- $W$  preconditioner (3.2). At  $x = 1/2$ , the logarithm disappears and the leading inverse is simply  $t = A/(\Lambda - C)$ .

The 2026 gamma-product formula of Arabi Ardehali and Rosengren [14] is a promising basis for making this inverse uniform when  $x$  itself varies with  $t$ , including transition scales not covered by a fixed Bernoulli jet.

### 9.4 Radial inversion at $q = -1$

Put  $q = -e^{-t}$ ,  $t \downarrow 0$ . Pairing even and odd factors gives the exact identity

$$(a; -e^{-t})_\infty = (a; e^{-2t})_\infty (-ae^{-t}; e^{-2t})_\infty. \quad (9.6)$$

Alternatively, split the Lambert series into even and odd indices. For  $|a| < 1$ , one obtains

$$\log(a; -e^{-t})_\infty = -\frac{\text{Li}_2(a^2)}{4t} + \frac{1}{2} \log(1-a) - \frac{a(a+3)}{12(1-a^2)} t + O(t^3). \quad (9.7)$$

The constant term simplifies nontrivially: the even-index contribution is  $\frac{1}{4} \log(1-a^2)$ , the odd-index contribution is  $-\frac{1}{4} \log((1+a)/(1-a))$ , and their sum is  $\frac{1}{2} \log(1-a)$ .

**Theorem 9.1** (PROVED Real radial inverse at  $-1$ ). *For real  $|a| < 1$ , let  $y = (a; -e^{-t})_\infty$ , and set*

$$A = \frac{\text{Li}_2(a^2)}{4}, \quad C = \frac{1}{2} \log(1-a), \quad D = \frac{a(a+3)}{12(1-a^2)}, \quad X = -\log y + C.$$

Then

$$t = \frac{A}{X} + \frac{DA^2}{X^3} + O(X^{-5}), \quad q = -e^{-t}. \quad (9.8)$$

*Proof.* Equation (9.7) has the canonical form of Equation (3.1). Formal reversion gives the result; standard remainder estimates for the Lambert series justify the asymptotic substitution on compact subsets of  $|a| < 1$ .  $\square$

## 9.5 A primitive-root inverse normal form

Let  $\zeta$  be primitive of order  $m$ , and approach radially through  $q = \zeta e^{-t}$ ,  $\Re t > 0$ . In

$$\log(a; q)_\infty = - \sum_{r \geq 1} \frac{a^r}{r(1 - q^r)},$$

only indices divisible by  $m$  produce the  $1/t$  singularity. Define

$$C_\zeta(a) = \frac{1}{2m} \log(1 - a^m) - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r}{r(1 - \zeta^r)}, \quad (9.9)$$

$$\mathcal{D}_\zeta(a) = \frac{a^m}{12(1 - a^m)} - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r \zeta^r}{(1 - \zeta^r)^2}. \quad (9.10)$$

Both series converge absolutely for  $|a| < 1$ .

**Theorem 9.2** (**PROVED** Radial cyclotomic inversion). *For  $|a| < 1$ ,*

$$\log(a; \zeta e^{-t})_\infty = - \frac{\text{Li}_2(a^m)}{m^2 t} + C_\zeta(a) - \mathcal{D}_\zeta(a)t + O(t^2). \quad (9.11)$$

Let

$$A = \frac{\text{Li}_2(a^m)}{m^2}, \quad X = -\log y + C_\zeta(a).$$

On a branch for which  $X \rightarrow \infty$ ,

$$t = \frac{A}{X} + \frac{\mathcal{D}_\zeta(a)A^2}{X^3} + O(X^{-4}), \quad q = \zeta e^{-t}. \quad (9.12)$$

*Proof.* For  $r = ms$ , expand

$$\frac{1}{1 - e^{-rt}} = \frac{1}{rt} + \frac{1}{2} + \frac{rt}{12} + O(t^3).$$

The singular sum is  $-\text{Li}_2(a^m)/(m^2 t)$ ; its constant and linear pieces give the first terms in (9.9) and (9.10). For  $m \nmid r$ ,

$$\frac{1}{1 - \zeta^r e^{-rt}} = \frac{1}{1 - \zeta^r} - \frac{r\zeta^r}{(1 - \zeta^r)^2} t + O(t^2).$$

Absolute convergence permits termwise summation. Reversion then follows as in (3.1).  $\square$

For  $m = 1$ ,  $C_1(a) = \frac{1}{2} \log(1 - a)$  and  $\mathcal{D}_1(a) = a/[12(1 - a)]$ . For  $m = 2$ , the formulas reduce exactly to (9.7). For  $m > 2$ , the constants are generally complex and encode a cyclic quantum-dilogarithm factor. Higher coefficients are convergent cyclotomic Lambert sums.

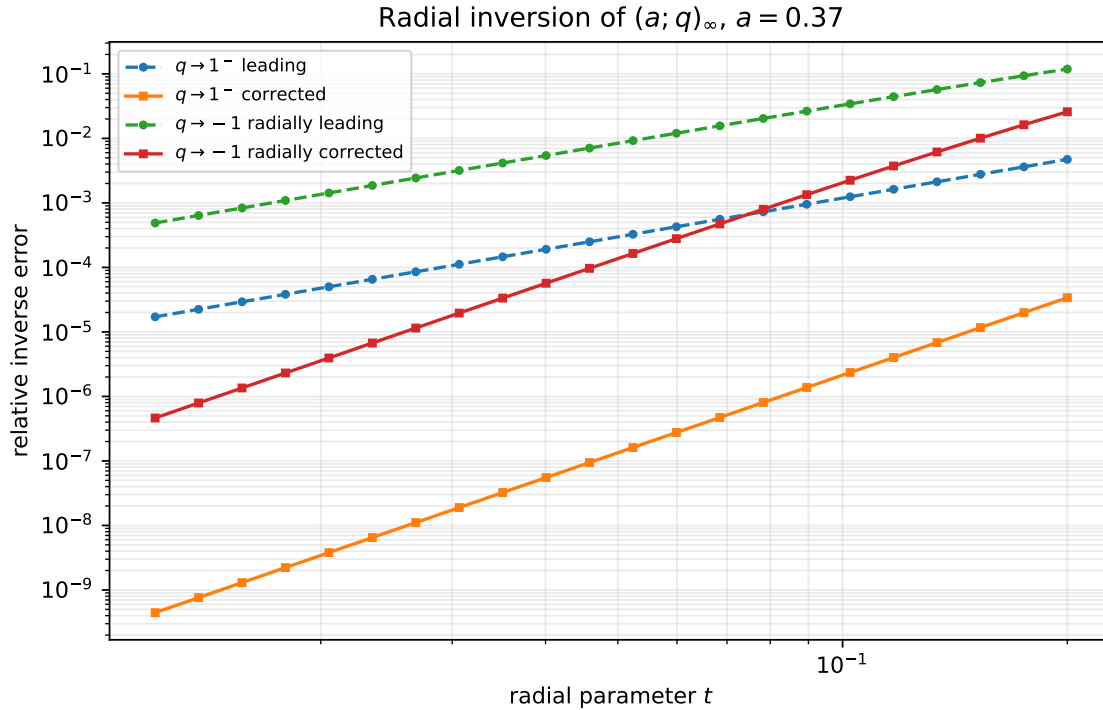


Figure 9.1: Relative error of the leading and corrected inverse formulas for  $(a; q)_\infty$ , with  $a = 0.37$ , on the rays  $q \rightarrow 1^-$  and  $q \rightarrow -1$ . The correction predicted by (3.1) changes the observed error order.

## 9.6 Exterior continuation and $q = \pm\infty$

The defining infinite product does not converge for  $|q| > 1$ . One must first choose an exterior normalization. Two common choices are

$$P^{\text{ext}}(a, q) = (a; q^{-1})_\infty, \quad |q| > 1,$$

or a  $q$ -gamma normalization in which explicit powers of  $q$  remove the divergent finite-product growth. For the first choice,  $Q = q^{-1} \rightarrow 0$ , and (9.1) applies directly after replacing  $q$  by  $Q$ . It is mathematically incorrect to discuss “the” value of  $(a; q)_\infty$  at  $q = \infty$  without specifying such a normalization.

## 9.7 Stokes data and continuation paths

The leading action  $A = \text{Li}_2(a^m)/m^2$  can rotate across a Stokes line, and  $C_\zeta(a)$  changes by logarithmic monodromy when  $a$  crosses the branch set of  $\text{Li}_2(a^m)$ . Consequently, two radial inverse expansions with the same formal powers may represent different analytic sheets. A robust complex algorithm stores the branch-state tuple

$$(\zeta, m, \arg t, \text{Log } y, \text{Log}(1 - a^m))$$

as part of every continuation record.

**Conjecture 9.3** (CONJECTURAL Cyclotomic resurgent inverse). After removal of the action  $\text{Li}_2(a^m)/(m^2t)$ , every radial branch of  $(a; q)_\infty$  admits a resurgent transseries whose perturbative coefficients belong to the differential algebra generated by cyclotomic Lambert sums, polylogarithms, and cyclic quantum dilogarithms. Its inverse transseries is closed under the same algebra, with Stokes automorphisms transported by formal composition.



## **Part III**

# **Gaussian, multinomial, and parameter inverses**

## Chapter 10

# Gaussian coefficients inverted in the base

### 10.1 The global positive branch

For  $0 < k < n$ , the Gaussian coefficient

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q$$

is a polynomial with nonnegative integer coefficients, constant term one, and degree

$$D = k(n - k).$$

It is palindromic:

$$G_{n,k}(q) = q^D G_{n,k}(q^{-1}). \quad (10.1)$$

The global positive-inverse theorem therefore applies.

**Theorem 10.1** (**PROVED** Unique positive Gaussian base). *For every  $0 < k < n$  and every  $y > 1$ , there exists exactly one  $q > 0$  such that  $\begin{bmatrix} n \\ k \end{bmatrix}_q = y$ . The solution depends analytically on  $y$  away from critical values in the complex extension.*

Although this theorem is elementary, it is algorithmically decisive: bisection on  $q > 0$  is globally safe, and a local asymptotic can be used only as an accelerator, never as an unverified branch selector.

### 10.2 The inverse at $q = 0$

The coefficient of  $q^j$  counts partitions of  $j$  contained in a  $k \times (n - k)$  rectangle. Write

$$G_{n,k}(q) = 1 + c_1 q + c_2 q^2 + c_3 q^3 + \cdots, \quad c_1 = 1.$$

Then, with  $u = y - 1$ ,

$$q = u - c_2 u^2 + (2c_2^2 - c_3) u^3 + O(u^4). \quad (10.2)$$

The second coefficient has the transparent form

$$c_2 = \mathbf{1}_{k \geq 2} + \mathbf{1}_{n-k \geq 2}.$$

More generally, any partition generator or dynamic program for the rectangle coefficients immediately becomes an all-order inverse generator through Bell reversion.

The inverse is badly conditioned for recovering  $n$  or  $k$  when  $q \rightarrow 0$ , because every nontrivial Gaussian coefficient tends to one. Base recovery remains well conditioned because the first coefficient is always one.

### 10.3 Bernoulli and cumulant reversion at $q = 1$

Set  $q = e^t$ , and define

$$P_r(n, k) = \sum_{j=1}^k ((n - k + j)^r - j^r).$$

The elementary expansion

$$\log \frac{e^x - 1}{x} = \frac{x}{2} + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} x^{2m}$$

gives

$$\log \frac{G_{n,k}(e^t)}{\binom{n}{k}} = \frac{D}{2}t + \sum_{m \geq 1} \frac{B_{2m} P_{2m}(n, k)}{2m(2m)!} t^{2m}. \quad (10.3)$$

This is an all-order expansion with no odd powers beyond the linear term. In probabilistic language, if  $J$  is the area of a uniformly selected partition in the rectangle, then

$$\mathbb{E}J = \frac{D}{2}, \quad \kappa_{2m}(J) = \frac{B_{2m}}{2m} P_{2m}(n, k), \quad \kappa_{2m+1}(J) = 0 \quad (m \geq 1).$$

In particular,

$$P_2 = D(n+1), \quad \text{Var } J = \frac{D(n+1)}{12}.$$

Let

$$s = \log \frac{y}{\binom{n}{k}}.$$

Reversion of the first two nonzero terms gives

$$t = \frac{2s}{D} - \frac{n+1}{3D^2} s^2 + \frac{(n+1)^2}{9D^3} s^3 + O(s^4), \quad q = e^t. \quad (10.4)$$

The coefficient of  $s^4$  receives the first contribution from the fourth cumulant; higher terms are generated mechanically.

*Remark 10.2.* Using  $t = \log q$ , rather than  $q - 1$ , exposes reciprocal symmetry and substantially simplifies the inverse. Conversion to a series in  $q - 1$  is best delayed until the final step.

### 10.4 The exact first inverse jets at $q = -1$

Write

$$n = 2a + r, \quad k = 2b + s, \quad r, s \in \{0, 1\}.$$

The  $q$ -Lucas theorem gives

$$G_{n,k}(-1) = \binom{a}{b} \binom{r}{s}. \quad (10.5)$$

Thus the value vanishes exactly when  $n$  is even and  $k$  is odd.

**Theorem 10.3** (PROVED Gaussian inverse at  $-1$ ). *If  $(r, s) \neq (0, 1)$ , put  $C = \binom{a}{b}$ . Then*

$$G_{n,k}(q) = C - \frac{D}{2}C(q+1) + O((q+1)^2), \quad (10.6)$$

and

$$q = -1 - \frac{2(y-C)}{DC} + O((y-C)^2).$$

If  $n = 2a$  and  $k = 2b + 1$ , then

$$G_{2a,2b+1}(q) = (a - b) \binom{a}{b} (q + 1) + O((q + 1)^2), \quad (10.7)$$

and

$$q = -1 + \frac{y}{(a - b) \binom{a}{b}} + O(y^2).$$

Every nontrivial Gaussian base inverse is therefore regular at  $q = -1$ , even when the forward value is zero.

*Proof.* The value formula is the order-two case of  $q$ -Lucas. In the nonzero case,  $D$  is even. Differentiate the reciprocal identity  $G(q) = q^D G(q^{-1})$  at  $q = -1$ ; the two derivative terms combine to give  $G'(-1) = -DG(-1)/2$ . In the zero case,  $q + 1 = \Phi_2(q)$  occurs with cyclotomic valuation one. Cancelling that factor in the product formula and taking the limit gives the coefficient  $(a - b) \binom{a}{b}$ . Ordinary reversion completes the proof.  $\square$

## 10.5 Primitive roots of unity and carry geometry

Let  $\zeta$  be primitive of order  $m$ . The exponent of  $\Phi_m(q)$  in a Gaussian coefficient is

$$\nu_m(n, k) = \lfloor n/m \rfloor - \lfloor k/m \rfloor - \lfloor (n - k)/m \rfloor. \quad (10.8)$$

This number is either zero or one. It equals one precisely when adding the least residues of  $k$  and  $n - k$  modulo  $m$  produces a carry. Equivalently, if

$$n = am + r, \quad k = bm + s, \quad 0 \leq r, s < m,$$

then a forced zero occurs exactly when  $s > r$ . The value in the no-carry case is

$$\begin{bmatrix} n \\ k \end{bmatrix}_\zeta = \binom{a}{b} \begin{bmatrix} r \\ s \end{bmatrix}_\zeta.$$

**Corollary 10.4** (**PROVED** Every forced cyclotomic zero is simple). *If  $\nu_m(n, k) = 1$ , then at each primitive  $m$ -th root  $\zeta$ ,*

$$G_{n,k}(q) = (q - \zeta) G'_{n,k}(\zeta) + O((q - \zeta)^2), \quad G'_{n,k}(\zeta) \neq 0.$$

*Hence the inverse near target zero is an ordinary Taylor branch, not a fractional-power branch.*

*Proof.* In  $(q; q)_r$ , the exponent of  $\Phi_m$  is  $\lfloor r/m \rfloor$ . Taking the difference in the Gaussian quotient gives (10.8). The floor identity restricts it to zero or one. Since cyclotomic polynomials have simple roots over characteristic zero, valuation one gives a nonzero first derivative.  $\square$

In the no-carry case,  $G(\zeta) \neq 0$ , but the derivative can in principle vanish. Such points are genuine complex critical points rather than forced cyclotomic zeros; they belong to the discriminant of  $G(q) - y$ .

## 10.6 The inverse at $q = \pm\infty$

Since  $c_1 = 1$ , (4.1) becomes

$$q = R^{-1} - \frac{1}{D} + \left( \frac{D-1}{2D^2} - \frac{c_2}{D} \right) R + O(R^2), \quad R = y^{-1/D}. \quad (10.9)$$

On the positive branch,  $R > 0$  and  $q \rightarrow +\infty$ . For real  $q \rightarrow -\infty$ , the sign of the target is  $(-1)^D$  to leading order. In the complex plane the  $D$  choices of  $R$  label the asymptotic inverse sheets.

## 10.7 Complex critical values and monodromy

The positive real branch has no critical points, but the complex polynomial usually has many. They are roots of  $G'_{n,k}(q)$ , and their images under  $G$  are the finite branch values. A computationally stable procedure is:

- (1) compute the polynomial by the Gaussian recurrence over exact integers;
- (2) isolate roots of  $G'$  with certified disks;
- (3) evaluate critical values with ball arithmetic;
- (4) connect roots of  $G(q) = y$  by homotopy while avoiding those disks.

A resultant gives the exact discriminant, but its coefficient growth makes it less useful than certified root tracking beyond modest degrees.

**Conjecture 10.5** (**CONJECTURAL** Generic Gaussian monodromy). After removing the reciprocal and exceptional cyclotomic symmetries, the monodromy group of a generic Gaussian polynomial  $G_{n,k}(q) - y$  is as large as allowed by those symmetries. Equivalently, almost all critical points are simple and the associated transpositions generate the full permitted permutation group of inverse sheets.

## Chapter 11

# Continuous upper and lower parameter inverses

### 11.1 The $q$ -gamma interpolation

For  $0 < q < 1$ , define

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}.$$

The continuous Gaussian coefficient is

$$\mathcal{B}_q(z, w) = \frac{\Gamma_q(z + 1)}{\Gamma_q(w + 1)\Gamma_q(z - w + 1)}. \quad (11.1)$$

It agrees with  $\begin{bmatrix} n \\ k \end{bmatrix}_q$  at integers. Its logarithmic derivatives are

$$\partial_z \log \mathcal{B}_q(z, w) = \psi_q(z + 1) - \psi_q(z - w + 1), \quad (11.2)$$

$$\partial_w \log \mathcal{B}_q(z, w) = -\psi_q(w + 1) + \psi_q(z - w + 1). \quad (11.3)$$

Here  $\psi_q = \Gamma'_q/\Gamma_q$ , and

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1 - q^{x+j})^2} > 0.$$

### 11.2 Upper-parameter inversion

For fixed  $w > 0$ , (11.2) is positive when  $z > w$ . Hence  $z \mapsto \mathcal{B}_q(z, w)$  is strictly increasing. At the lower endpoint  $z = w$ , the value is one. As  $z \rightarrow \infty$ ,

$$\mathcal{B}_q(z, w) \longrightarrow L_q(w) = \frac{(q^{w+1}; q)_\infty}{(q; q)_\infty}. \quad (11.4)$$

**Theorem 11.1** (PROVED Upper inverse for  $0 < q < 1$ ). *For fixed  $w > 0$  and every  $1 < y < L_q(w)$ , there is exactly one real  $z > w$  satisfying  $\mathcal{B}_q(z, w) = y$ . No finite real solution exists for  $y \geq L_q(w)$ .*

*Proof.* Strict monotonicity follows from the increasing  $q$ -digamma function. The endpoint values follow from (11.1) and the asymptotic cancellation of  $(q^z; q)_\infty$ .  $\square$

At  $z = w$ , put

$$A = \psi_q(w + 1) - \psi_q(1) > 0.$$

Then

$$z - w = \frac{y - 1}{A} + O((y - 1)^2).$$

At the upper saturation, let  $z \rightarrow \infty$ . Expanding the ratio of tail products gives, for integer  $w = k$ ,

$$L_q(k) - \mathcal{B}_q(z, k) \sim L_q(k)[k]_q q^{z-k+1}. \quad (11.5)$$

Therefore

$$z \sim k - 1 + \log_q \left( \frac{L_q(k) - y}{L_q(k)[k]_q} \right). \quad (11.6)$$

The logarithmic divergence explains why upper-parameter recovery becomes ill-conditioned near saturation.

For  $q > 1$ , reciprocal gamma normalization gives

$$\mathcal{B}_q(z, w) = q^{w(z-w)} \mathcal{B}_{1/q}(z, w). \quad (11.7)$$

Thus, for fixed  $w$ , the upper inverse grows logarithmically with the target:

$$z = w + \frac{\log y + \log(q^{-1}; q^{-1})_w}{w \log q} + o(1)$$

for integer  $w$ , with the obvious gamma-product replacement for continuous  $w$ .

### 11.3 Lower-parameter inversion and the central fold

Equation (11.3) is positive for  $w < z/2$ , zero at  $w = z/2$ , and negative for  $w > z/2$ . Hence  $w \mapsto \mathcal{B}_q(z, w)$  is symmetric and has a unique maximum

$$M_q(z) = \mathcal{B}_q(z, z/2).$$

**Theorem 11.2** (PROVED Two lower-parameter sheets). *For fixed  $z > 0$ , every target  $1 < y < M_q(z)$  has exactly two real solutions  $w_- < z/2 < w_+$ , related by  $w_+ = z - w_-$ . Near the maximum,*

$$w_{\pm} = \frac{z}{2} \pm \sqrt{\frac{M_q(z) - y}{M_q(z)\psi'_q(z/2 + 1)}} + O(M_q(z) - y). \quad (11.8)$$

*Proof.* Monotonicity and symmetry give the global branch count. At the center,

$$\partial_w^2 \log \mathcal{B}_q(z, w)|_{w=z/2} = -2\psi'_q(z/2 + 1),$$

so  $\mathcal{B}_q''(z, z/2) = -2M_q(z)\psi'_q(z/2 + 1)$ . The fold formula gives (11.8).  $\square$

Near  $w = 0$ , let

$$A_z = \psi_q(z + 1) - \psi_q(1).$$

Then

$$w = \frac{y - 1}{A_z} - \frac{A_z^2 - \psi'_q(1) - \psi'_q(z + 1)}{2A_z^3}(y - 1)^2 + O((y - 1)^3).$$

This branch is useful when a Gaussian coefficient is continued from a sparse lower index.

## 11.4 Discrete aliasing at special bases

At  $q = 0$ , every nontrivial integer Gaussian coefficient equals one, so integer upper/lower parameters cannot be recovered from a single value. At  $q = -1$ ,  $q$ -Lucas reduces the value to a smaller ordinary binomial plus a parity bit; many distinct  $(n, k)$  pairs alias. At a primitive  $m$ -th root, the residues modulo  $m$  and the quotient-level binomial are the recoverable data. Parameter inversion at roots of unity is therefore intrinsically arithmetic and set-valued.

## 11.5 A dilogarithmic double-scaling inverse

Let

$$q = e^{-\tau/n}, \quad k = \rho n, \quad 0 < \rho < 1,$$

with  $n \rightarrow \infty$ . Euler–Maclaurin applied to the product gives

$$\frac{1}{n} \log \left[ \frac{n}{\rho n} \right]_q \longrightarrow \Phi(\rho, \tau), \quad (11.9)$$

where

$$\Phi(\rho, \tau) = \frac{1}{\tau} \left[ \frac{\pi^2}{6} + \text{Li}_2(e^{-\tau}) - \text{Li}_2(e^{-\rho\tau}) - \text{Li}_2(e^{-(1-\rho)\tau}) \right]. \quad (11.10)$$

The rate is symmetric under  $\rho \leftrightarrow 1 - \rho$  and has a unique maximum at  $\rho = 1/2$ . Consequently, inversion in  $\rho$  has two branches and a central square-root transition, while inversion in  $\tau$  is governed by derivatives of dilogarithms. Finite-size corrections can be reverted jointly with (11.10) to recover  $(n, \rho)$  from several observables.

**Research problem 11.3.** Develop a uniform inverse expansion through the three simultaneous limits  $n \rightarrow \infty$ ,  $q \rightarrow 1$ , and  $\rho \rightarrow 1/2$ . The expected normal form combines an error-function central scaling with dilogarithmic large deviations and should provide a  $q$ -analogue of uniform inverse-binomial asymptotics.



## Chapter 12

# Multinomials and broader positive $q$ -analogs

### 12.1 Gaussian multinomials

For nonnegative integers  $k_1 + \cdots + k_r = n$ , define

$$\left[ \begin{matrix} n \\ k_1, \dots, k_r \end{matrix} \right]_q = \frac{(q; q)_n}{\prod_{i=1}^r (q; q)_{k_i}}.$$

It is a positive palindromic polynomial of degree

$$D = \sum_{1 \leq i < j \leq r} k_i k_j.$$

Thus it has a unique positive base inverse for every  $y > 1$ , the universal small- $q$  inverse, and the reciprocal large- $q$  inverse.

At  $q = e^t$ ,

$$\log \frac{\left[ \begin{matrix} n \\ k_1, \dots, k_r \end{matrix} \right]_q}{\left( \begin{matrix} n \\ k_1, \dots, k_r \end{matrix} \right)} = \frac{D}{2} t \tag{12.1}$$

$$+ \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} \left( \sum_{j=1}^n j^{2m} - \sum_{i=1}^r \sum_{j=1}^{k_i} j^{2m} \right) t^{2m}. \tag{12.2}$$

Hence all base-inverse coefficients are explicit polynomials in power sums and Bernoulli numbers.

### 12.2 A reusable inversion template

Many  $q$ -Catalan,  $q$ -Narayana, rank-generating, Hilbert, and Poincaré polynomials satisfy

$$P(q) = \sum c_j q^j, \quad c_j \geq 0, \quad P(q) = q^D P(q^{-1}).$$

For every such object:

- (i) positive-base inversion is globally unique;
- (ii) the  $q = 0$  inverse is governed by the first combinatorial coefficients;

- (iii) the  $q = 1$  inverse is cumulant reversion for the graded statistic;
- (iv) the  $q = \infty$  inverse is determined by reciprocity;
- (v) roots of unity are controlled by cyclotomic valuations and cancellations.

This turns a large class of apparently unrelated inverse problems into one standard pipeline.

### 12.3 Coefficient distributions and asymptotic inverse universality

Suppose a sequence  $P_N(q)$  has exponent variables  $J_N$  whose centered cumulants, after scaling, approach those of a Gaussian. Then near  $q = 1$ ,

$$\log \frac{P_N(e^t)}{P_N(1)} = \mu_N t + \frac{\sigma_N^2 t^2}{2} + o(\sigma_N^2 t^2)$$

uniformly on a shrinking window. The inverse is then universally

$$t = \frac{s}{\mu_N} - \frac{\sigma_N^2}{2\mu_N^3} s^2 + \dots$$

Non-Gaussian limiting cumulants produce systematic corrections. This probabilistic organization is particularly effective for Gaussian coefficients, plane partitions, and graded representation dimensions.

**Conjecture 12.1** (**CONJECTURAL** Inverse cumulant universality). For broad mod-Gaussian families of positive palindromic  $q$ -polynomials, the base inverse admits a uniform expansion in standardized cumulants whose domain extends up to the nearest complex saddle of the limiting cumulant-generating function. The first failure of the expansion is governed by a universal Airy-type coalescence of inverse saddles.

## **Part IV**

# **Other $q$ -analogs and their parameter inverses**

## Chapter 13

# Inverse $q$ -numbers and $q$ -factorials

### 13.1 The $q$ -number inverted in $x$

For fixed  $q \notin \{0, 1\}$ ,

$$[x]_q = \frac{1 - q^x}{1 - q} = y$$

is elementary to invert, but multivalued over  $\mathbb{C}$ :

$$x = \frac{\text{Log}(1 - (1 - q)y) + 2\pi i k}{\text{Log } q}, \quad k \in \mathbb{Z}. \quad (13.1)$$

The denominator also depends on the chosen logarithm of  $q$ . For real  $q > 0$ ,  $q \neq 1$ , and a positive logarithm branch, the real inverse is unique on every interval where the target remains in range.

At  $q = 1$ ,  $[x]_1 = x$ , so inversion is the identity. At  $q = 0$ , the continuation depends sharply on  $\Re x$ : for  $\Re x > 0$ ,  $[x]_0 = 1$ , and information about  $x$  is lost; for  $\Re x < 0$ , the term  $q^x$  diverges.

### 13.2 The base inverse of $[x]_q$

For fixed  $x$ , the base equation is

$$q^x - yq + (y - 1) = 0. \quad (13.2)$$

It is a polynomial when  $x$  is a nonnegative integer and a transcendental, branch-dependent equation otherwise.

At  $q = e^t$ ,

$$\frac{[x]_{e^t}}{x} = e^{(x-1)t/2} \frac{\sinh(xt/2)/(xt/2)}{\sinh(t/2)/(t/2)}. \quad (13.3)$$

Consequently,

$$\log \frac{[x]_{e^t}}{x} = \frac{x-1}{2}t + \sum_{m \geq 1} \frac{B_{2m}(x^{2m} - 1)}{2m(2m)!} t^{2m}. \quad (13.4)$$

For  $x \neq 0, 1$ , set  $s = \log(y/x)$ . Then

$$t = \frac{2s}{x-1} - \frac{x+1}{3(x-1)^2} s^2 + \frac{(x+1)^2}{9(x-1)^3} s^3 + O(s^4), \quad q = e^t. \quad (13.5)$$

The values  $x = 0$  and  $x = 1$  are constant maps and have no base inverse.

### 13.3 Fractional Newton polygons at $q = 0$

For noninteger  $x > 0$ ,

$$[x]_q = \frac{1 - q^x}{1 - q} = 1 + q + q^2 + \cdots - q^x(1 + q + \cdots).$$

There are three basic inverse regimes.

(i) If  $0 < x < 1$ , then  $1 - y \sim q^x$ , so

$$q \sim (1 - y)^{1/x}.$$

The next exponent is selected by comparing  $2x$  and 1.

(ii) If  $x > 1$ , then  $y - 1 \sim q$ , but a fractional correction  $q^x$  enters before  $q^2$  when  $1 < x < 2$ .

(iii) If  $x$  is a positive integer, the expansion is an ordinary polynomial inverse and coefficient positivity gives the unique positive branch for  $y > 1$ .

These are elementary examples of the generalized-series mechanism developed in [Section 3.2](#).

### 13.4 The point $q = -1$

For integer  $n \geq 2$ ,

$$[2m]_{-1} = 0, \quad [2m + 1]_{-1} = 1.$$

Taylor expansion gives

$$[2m]_q = m(q + 1) + O((q + 1)^2), \quad (13.6)$$

$$[2m + 1]_q = 1 - m(q + 1) + O((q + 1)^2). \quad (13.7)$$

Hence

$$q = -1 + \frac{y}{m} + O(y^2)$$

for even order, and

$$q = -1 + \frac{1 - y}{m} + O((1 - y)^2)$$

for odd order. For noninteger  $x$ , the value  $(-1)^x$  depends on a logarithm branch, so  $q = -1$  is itself a branch point in the base variable.

### 13.5 The $q$ -factorial near $q = 0$

Let

$$F_n(q) = [n]_q! = \prod_{j=1}^n [j]_q.$$

For  $n \geq 2$ ,

$$F_n(q) = 1 + (n - 1)q + \frac{(n - 2)(n + 1)}{2}q^2 + O(q^3).$$

Thus

$$q = \frac{y - 1}{n - 1} - \frac{(n - 2)(n + 1)}{2(n - 1)^3}(y - 1)^2 + O((y - 1)^3). \quad (13.8)$$

Since all coefficients are positive, this is the local form of the unique positive base inverse.

### 13.6 The $q$ -factorial near $q = 1$

The degree is

$$N = \binom{n}{2}.$$

Summing (13.4) over  $x = 1, \dots, n$  gives

$$\log \frac{[n]_{e^t}!}{n!} = \frac{N}{2}t + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} \sum_{j=1}^n (j^{2m} - 1)t^{2m}. \quad (13.9)$$

Since

$$\sum_{j=1}^n (j^2 - 1) = \frac{N(2n + 5)}{3},$$

with  $s = \log(y/n!)$ ,

$$t = \frac{2s}{N} - \frac{2n + 5}{9N^2}s^2 + \frac{(2n + 5)^2}{81N^3}s^3 + O(s^4). \quad (13.10)$$

### 13.7 Cyclotomic Puiseux inverses of the factorial

Let  $\zeta \neq 1$  be primitive of order  $m$ , and put

$$\nu = \lfloor n/m \rfloor.$$

Exactly the factors  $[m]_q, [2m]_q, \dots, [\nu m]_q$  vanish at  $\zeta$ , each simply. Therefore

$$[n]_q! = C_{n,m,\zeta}(q - \zeta)^\nu (1 + O(q - \zeta)), \quad (13.11)$$

where

$$C_{n,m,\zeta} = \prod_{\substack{1 \leq j \leq n \\ m \nmid j}} [j]_\zeta \prod_{r=1}^{\nu} \frac{-mr\zeta^{-1}}{1 - \zeta}.$$

The inverse at target zero has  $\nu$  sheets:

$$q - \zeta \sim \left( \frac{y}{C_{n,m,\zeta}} \right)^{1/\nu}.$$

At  $m = 2, \zeta = -1$ , write  $n = 2\nu$  or  $2\nu + 1$ . A more precise real expansion is

$$[2\nu]_q! = \nu!(q + 1)^\nu (1 - \nu(\nu - 1)(q + 1) + O((q + 1)^2)), \quad (13.12)$$

$$[2\nu + 1]_q! = \nu!(q + 1)^\nu (1 - \nu^2(q + 1) + O((q + 1)^2)). \quad (13.13)$$

With  $\rho = (y/\nu!)^{1/\nu}$ , the corresponding sheets satisfy

$$q + 1 = \rho + (\nu - 1)\rho^2 + O(\rho^3), \quad n = 2\nu, \quad (13.14)$$

$$q + 1 = \rho + \nu\rho^2 + O(\rho^3), \quad n = 2\nu + 1. \quad (13.15)$$

### 13.8 The factorial inverse at infinity

Reciprocity gives

$$[n]_q! = q^N [n]_{q^{-1}}!.$$

Since the coefficient of  $q$  at the origin is  $n - 1$ ,

$$q = y^{1/N} - \frac{n - 1}{N} + O(y^{-1/N}) = y^{1/N} - \frac{2}{n} + O(y^{-1/N}). \quad (13.16)$$

The same formula, with a selected  $N$ -th root, describes all complex asymptotic sheets.

## Chapter 14

# Inverse $q$ -gamma and $q$ -digamma functions

### 14.1 Forward formulas

For  $0 < q < 1$ ,

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}, \quad (14.1)$$

$$\psi_q(x) = -\log(1 - q) + (\log q) \sum_{j \geq 0} \frac{q^{x+j}}{1 - q^{x+j}}, \quad (14.2)$$

$$\psi'_q(x) = (\log q)^2 \sum_{j \geq 0} \frac{q^{x+j}}{(1 - q^{x+j})^2} > 0. \quad (14.3)$$

The monotonicity consequences agree with the standard theory summarized in the DLMF [17]. Moak's  $q$ -Stirling expansion controls large  $x$  uniformly in important regimes [8].

### 14.2 Inverting $\Gamma_q$ in its argument

Because  $\psi'_q > 0$ ,  $\psi_q$  has at most one zero. As  $x \downarrow 0$ ,  $\psi_q(x) \rightarrow -\infty$ ; as  $x \rightarrow \infty$ ,  $\psi_q(x) \rightarrow -\log(1 - q) > 0$ . Hence there is a unique  $x_q^* > 0$  at which  $\Gamma_q$  attains its minimum

$$M_q = \Gamma_q(x_q^*).$$

**Theorem 14.1** (PROVED Real branch count for inverse  $q$ -gamma). *For  $y > M_q$ , the equation  $\Gamma_q(x) = y$  has exactly two positive real solutions; for  $y = M_q$ , one double solution; for  $0 < y < M_q$ , no positive real solution. Near the minimum,*

$$x = x_q^* \pm \sqrt{\frac{2(y - M_q)}{M_q \psi'_q(x_q^*)}} + O(y - M_q). \quad (14.4)$$

*Proof.* Strict increase of  $\psi_q$  makes  $\Gamma_q$  decrease before its unique critical point and increase afterward. At the critical point,  $\Gamma''_q = M_q \psi'_q(x_q^*) > 0$ , and fold inversion gives the expansion.  $\square$

### 14.3 Left and right endpoint inverses in $x$

Using the recurrence  $\Gamma_q(x + 1) = [x]_q \Gamma_q(x)$ , one finds

$$\log \Gamma_q(x) = -\log x + \log \frac{1 - q}{-\log q} + c_q x + O(x^2), \quad c_q = \psi_q(1) - \frac{1}{2} \log q. \quad (14.5)$$

Therefore, as  $y \rightarrow \infty$  on the left branch,

$$x = \frac{A_q}{y} + \frac{A_q^2 c_q}{y^2} + O(y^{-3}), \quad A_q = \frac{1-q}{-\log q}. \quad (14.6)$$

On the right branch,

$$\Gamma_q(x) = C_q(1-q)^{-x} \left( 1 + \frac{q^x}{1-q} + O(q^{2x}) \right), \quad C_q = (q; q)_\infty(1-q).$$

Let

$$x_0 = \frac{\log y - \log C_q}{-\log(1-q)}.$$

Then

$$x = x_0 - \frac{q^{x_0}}{(1-q)(-\log(1-q))} + O(q^{2x_0}). \quad (14.7)$$

## 14.4 Base inversion near $q = 0$

From (14.1), one obtains the generalized expansion

$$\log \Gamma_q(x) = q^x + (x-2)q + \frac{1}{2}q^{2x} + q^{x+1} + \frac{x-4}{2}q^2 + \cdots, \quad (14.8)$$

where terms must be sorted by exponent and combined when exponents collide. This immediately gives distinct inverse regimes for  $x > 0$ :

(i)  $0 < x < 1$ : if  $s = \log y \downarrow 0$ , then

$$q \sim s^{1/x}.$$

The next term comes from  $q^{2x}$  for  $x < 1/2$ , from  $q$  for  $x > 1/2$ , and from both at  $x = 1/2$ .

(ii)  $1 < x < 2$ :  $s \sim (x-2)q < 0$ , so

$$q \sim \frac{s}{x-2}.$$

(iii)  $x > 2$ : the same formula holds with  $s > 0$ .

(iv)  $x = 1$  and  $x = 2$ :  $\Gamma_q(x) = 1$  identically, so base recovery from this observable is impossible.

This is a clean example where ordinary Taylor reversion fails solely because a continuous parameter creates fractional exponents.



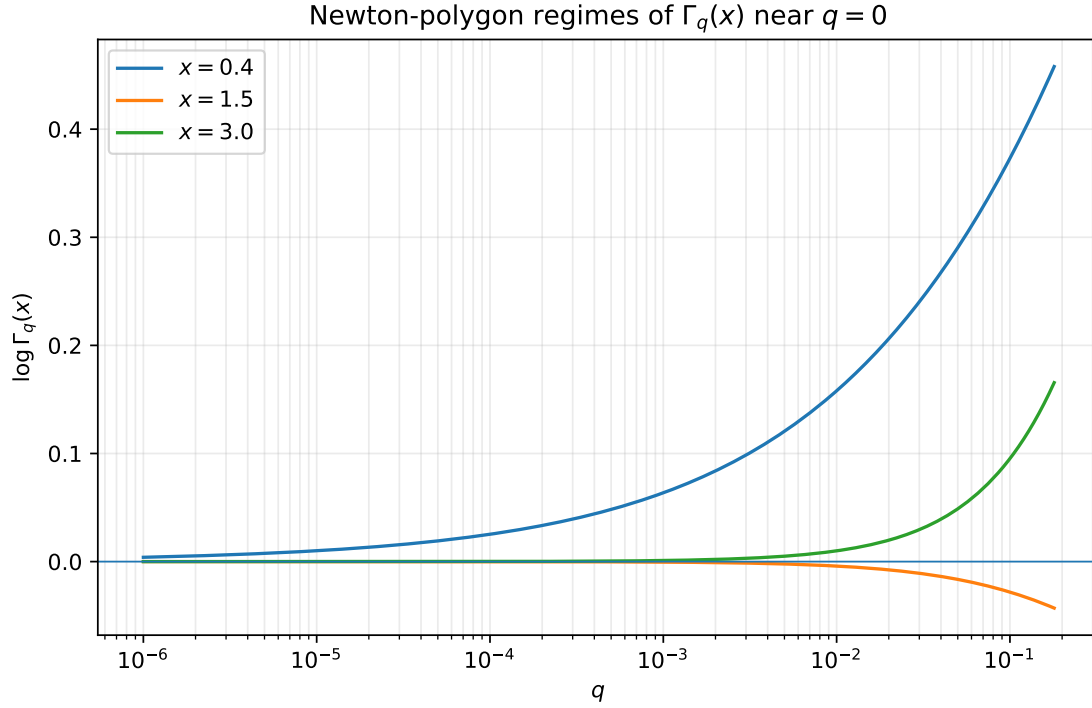


Figure 14.1: The three generic small-base regimes of  $\log \Gamma_q(x)$ . For  $0 < x < 1$ , the fractional term  $q^x$  dominates; for  $1 < x < 2$  and  $x > 2$ , the linear term has opposite signs.

### 14.5 Base inversion near $q = 1$

Combining the Bernoulli expansion of  $(q^x; q)_\infty$  with the eta-product asymptotic gives, for  $q = e^{-t}$ ,

$$\log \frac{\Gamma_{e^{-t}}(x)}{\Gamma(x)} = -A_x t + B_x t^2 + O(t^3), \quad (14.9)$$

where

$$A_x = \frac{(x-1)(x-2)}{4}, \quad B_x = \frac{(x-1)(x-2)(2x+3)}{144}. \quad (14.10)$$

The sign of  $A_x$  reproduces the known change of monotonicity across  $x = 1$  and  $x = 2$ .

For  $x \notin \{1, 2\}$ , set  $s = \log(y/\Gamma(x))$ . Then

$$t = -\frac{s}{A_x} + \frac{B_x}{A_x^3} s^2 - \frac{2B_x^2}{A_x^5} s^3 + O(s^4), \quad q = e^{-t}. \quad (14.11)$$

At  $x = 1, 2$ , every coefficient vanishes because the function is exactly constant, not merely critical to finite order.

### 14.6 The exterior base and $q \rightarrow \infty$

For  $q > 1$ , the standard normalization satisfies

$$\Gamma_q(x) = q^{(x-1)(x-2)/2} \Gamma_{1/q}(x). \quad (14.12)$$

Thus, away from  $x = 1, 2$ , the leading base inverse is the power law

$$q \sim y^{2/((x-1)(x-2))}, \quad (14.13)$$

with the interpretation reversed when the exponent is negative. Corrections are obtained by substituting the small- $Q$  generalized expansion (14.8),  $Q = q^{-1}$ , and reverting. For complex or negative  $q$ , a branch of  $q^{(x-1)(x-2)/2}$  must be specified.

### 14.7 Inverse $q$ -digamma in $x$

For fixed  $0 < q < 1$ ,  $\psi_q$  is a strictly increasing bijection

$$\psi_q : (0, \infty) \longrightarrow (-\infty, -\log(1-q)).$$

Hence its real inverse exists exactly on that interval. From (14.5),

$$\psi_q(x) = -\frac{1}{x} + c_q + O(x),$$

so for  $y \rightarrow -\infty$ ,

$$x = -\frac{1}{y - c_q} + O(y^{-3}). \quad (14.14)$$

At the upper endpoint,

$$-\log(1-q) - \psi_q(x) \sim \frac{-\log q}{1-q} q^x,$$

and therefore

$$x \sim \frac{\log(-\log(1-q) - y) + \log((1-q)/(-\log q))}{\log q}. \quad (14.15)$$

### 14.8 The critical base inverse of $\psi_q$

Differentiating (14.9) in  $x$  yields

$$\psi_{e^{-t}}(x) - \psi(x) = -\frac{2x-3}{4}t + \frac{6x^2-6x-5}{144}t^2 + O(t^3). \quad (14.16)$$

For  $x \neq 3/2$ , base inversion is regular. At  $x = 3/2$ , the linear term vanishes and

$$\psi_{e^{-t}}(3/2) = \psi(3/2) - \frac{t^2}{288} + O(t^3). \quad (14.17)$$

Thus the base inverse changes to a square-root branch:

$$t \sim \sqrt{288(\psi(3/2) - y)}.$$

This critical value is the derivative-in- $x$  counterpart of the fixed points  $x = 1, 2$  for  $\Gamma_q$ .

# Chapter 15

## Inverse $q$ -beta functions

### 15.1 Monotone parameter inverses

Define

$$B_q(x, y) = \frac{\Gamma_q(x)\Gamma_q(y)}{\Gamma_q(x+y)}.$$

For positive  $x, y$ ,

$$\begin{aligned}\partial_x \log B_q(x, y) &= \psi_q(x) - \psi_q(x+y) < 0, \\ \partial_y \log B_q(x, y) &= \psi_q(y) - \psi_q(x+y) < 0.\end{aligned}$$

Thus either shape parameter has a unique real inverse when the other is fixed and the target is in range.

As  $x \downarrow 0$ ,

$$B_q(x, y) \sim \frac{1-q}{(-\log q)x}.$$

As  $x \rightarrow \infty$ ,

$$B_q(x, y) \longrightarrow S_q(y) = \Gamma_q(y)(1-q)^y = \frac{(q; q)_\infty(1-q)}{(q^y; q)_\infty}. \quad (15.1)$$

Therefore the real  $x$ -inverse exists for targets above  $S_q(y)$ . Near saturation,

$$B_q(x, y) = S_q(y) (1 + [y]_q q^x + O(q^{2x})),$$

so

$$x \sim \log_q \left( \frac{B_q(x, y)/S_q(y) - 1}{[y]_q} \right). \quad (15.2)$$

### 15.2 Base inversion near $q = 1$

Using (14.9), one finds

$$\log \frac{B_{e^{-t}}(x, y)}{B(x, y)} = \frac{xy-1}{2}t - \frac{x^2y + xy^2 - xy - 1}{24}t^2 + O(t^3). \quad (15.3)$$

The linear coefficient vanishes on the hyperbola  $xy = 1$ .

**Theorem 15.1** (PROVED Critical curve for inverse  $q$ -beta). *If  $xy \neq 1$ , the base inverse near  $q = 1$  is an ordinary Taylor branch. If  $xy = 1$  but  $(x, y) \neq (1, 1)$ , then*

$$\log \frac{B_{e^{-t}}(x, y)}{B(x, y)} = -\frac{x+y-2}{24}t^2 + O(t^3), \quad (15.4)$$

and

$$t \sim \sqrt{\frac{-24 \log(B_{e^{-t}}(x, y)/B(x, y))}{x + y - 2}}. \quad (15.5)$$

$At(x, y) = (1, 1)$ ,  $B_q(1, 1) = 1$  *identically and base recovery is impossible*.

*Proof.* Substitute  $xy = 1$  into the quadratic coefficient of (15.3); it becomes  $-(x + y - 2)/24$ . Since  $x + y \geq 2$  for positive reciprocal variables, the coefficient is nonzero except at  $x = y = 1$ . Regular or fold reversion then applies.  $\square$

For  $xy \neq 1$ , if

$$s = \log \frac{y_0}{B(x, y)}, \quad A = \frac{xy - 1}{2}, \quad C = -\frac{x^2y + xy^2 - xy - 1}{24},$$

then

$$t = \frac{s}{A} - \frac{C}{A^3}s^2 + O(s^3).$$

### 15.3 The exterior inverse

The reciprocity of  $q$ -gamma gives

$$B_q(x, y) = q^{1-xy} B_{1/q}(x, y), \quad q > 1. \quad (15.6)$$

Hence, for  $xy \neq 1$ ,

$$q \sim \left( \frac{y_0}{B_0(x, y)} \right)^{1/(1-xy)}, \quad (15.7)$$

where  $B_0(x, y) = 1$  in the generic positive-parameter normalization, subject to the fractional small- $q$  corrections inherited from (14.8). The surface  $xy = 1$  is again exceptional: the explicit power vanishes and the first inverse information comes from the small- $1/q$  correction.

## Chapter 16

# Euler $q$ -exponentials and basic hypergeometric inverses

### 16.1 Exact reduction of Euler exponentials

Use the unscaled Euler functions

$$e_q(z) = \frac{1}{(z; q)_\infty}, \quad E_q(z) = (-z; q)_\infty, \quad |q| < 1.$$

Their argument inverses are exactly the argument inverse of the infinite product:

$$e_q^{-1}(y) = \mathcal{A}_q(1/y), \quad E_q^{-1}(y) = -\mathcal{A}_q(y), \quad (16.1)$$

with matching sheet labels. Every endpoint expansion in the infinite-product chapters transfers immediately. The scaled convention  $e_q((1-q)z) \rightarrow e^z$  merely adds a factor  $(1-q)^{-1}$  after inversion.

Base inversion likewise reduces to

$$(z; q)_\infty = 1/y \quad \text{or} \quad (-z; q)_\infty = y.$$

Thus the radial formulas at  $q = 1, -1$ , and primitive roots are already a complete base-inverse asymptotic theory for Euler  $q$ -exponentials.

### 16.2 Local inverse of a basic hypergeometric series in its argument

Adopt the standard convention

$$\begin{aligned} {}_r\phi_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z \right) &= \sum_{n \geq 0} c_n z^n, \\ c_n &= \frac{(a_1, \dots, a_r; q)_n}{(q, b_1, \dots, b_s; q)_n} \left( (-1)^n q^{\binom{n}{2}} \right)^{1+s-r}. \end{aligned} \quad (16.2)$$

If  $c_1 \neq 0$ , the local inverse at target one is

$$z = \frac{y-1}{c_1} - \frac{c_2}{c_1^3}(y-1)^2 + \frac{2c_2^2 - c_1c_3}{c_1^5}(y-1)^3 + O((y-1)^4). \quad (16.3)$$

This formula covers  $q$ -Bessel,  $q$ -Airy, and many orthogonal-polynomial generating functions after specialization.

If  $c_1 = 0$ , one must locate the first nonzero  $c_r$ . The inverse then has  $r$  Puiseux sheets. Vanishing may be caused by a numerator parameter equal to one, by a root-of-unity base, or by cancellation between several parameters.

### 16.3 Inversion in a numerator parameter

Fix all variables except  $a_1$ . Near  $a_1 = 1$ ,

$$(a_1; q)_n = (1 - a_1)(a_1 q; q)_{n-1}, \quad n \geq 1.$$

Thus the entire nonconstant part of the series turns on linearly:

$${}_r\phi_s = 1 + (1 - a_1) \mathcal{S}(q, z, \mathbf{a}', \mathbf{b}) + O((1 - a_1)^2), \quad (16.4)$$

where  $\mathcal{S}$  is an explicitly convergent basic hypergeometric sum formed by replacing  $(a_1; q)_n/(1 - a_1)$  with  $(q; q)_{n-1}$  at  $a_1 = 1$ . If  $\mathcal{S} \neq 0$ ,

$$a_1 = 1 - \frac{y - 1}{\mathcal{S}} + O((y - 1)^2).$$

At  $a_1 = q^{-N}$ , the series terminates. Moving the parameter away from this point activates the tail  $n > N$  linearly, so the inverse is regular unless a second cancellation occurs. Near a denominator parameter  $b_j = q^{-N}$ , poles replace zeros and reciprocal target coordinates are more appropriate.

### 16.4 Inversion in the base

For fixed  $z$  inside the convergence disk, each coefficient in (16.2) has a finite or infinite  $q$ -Pochhammer expansion. Near  $q = 0$ , a Newton polygon in the parameter exponents gives the inverse. Near  $q = 1$ , one first rescales parameters as  $a_i = q^{\alpha_i}$ ,  $b_j = q^{\beta_j}$  when a nontrivial classical hypergeometric limit is desired. Fixed parameters and power-scaled parameters are different asymptotic problems and should not be mixed.

At roots of unity, individual coefficients may have poles while the complete series remains finite after cancellation. The correct order of operations is:

- (1) combine terms or use a transformation identity that makes cancellation explicit;
- (2) extract the radial action and cyclic factor;
- (3) only then revert the target relation.

Termwise inversion before cancellation is generally wrong.

### 16.5 Implicit eigenvalue and zero inverses

Many  $q$ -orthogonal polynomials satisfy a three-term recurrence

$$p_{n+1}(x; \lambda) = (x - \alpha_n(\lambda))p_n(x; \lambda) - \beta_n(\lambda)p_{n-1}(x; \lambda),$$

where  $\lambda$  may be  $q$  or a shape parameter. A simple zero  $x_j(\lambda)$  obeys

$$\frac{dx_j}{d\lambda} = -\frac{\partial_\lambda p_n(x_j; \lambda)}{\partial_x p_n(x_j; \lambda)}.$$

This is the same partial-inverse sensitivity formula as before. At a multiple zero, eigenvalue sheets coalesce and Puiseux expansions replace Taylor series. The Jacobi-matrix representation supplies certified real brackets and a stable continuation algorithm.

# Chapter 17

## Theta functions and the inverse nome

### 17.1 Small-nome inverse series

For the standard theta constants,

$$\begin{aligned}\vartheta_2(0, q) &= 2q^{1/4}(1 + q^2 + q^6 + \cdots), \\ \vartheta_3(0, q) &= 1 + 2q + 2q^4 + 2q^9 + \cdots, \\ \vartheta_4(0, q) &= 1 - 2q + 2q^4 - 2q^9 + \cdots.\end{aligned}$$

Hence, on the principal small positive branch,

$$q = \left(\frac{y}{2}\right)^4 \left(1 - 4\left(\frac{y}{2}\right)^8 + O(y^{16})\right), \quad y = \vartheta_2(0, q), \quad (17.1)$$

$$q = u - u^4 + O(u^7), \quad u = \frac{\vartheta_3(0, q) - 1}{2}, \quad (17.2)$$

$$q = v + v^4 + O(v^7), \quad v = \frac{1 - \vartheta_4(0, q)}{2}. \quad (17.3)$$

The fractional fourth power in  $\vartheta_2$  is a genuine four-sheeted complex inverse at the origin.

### 17.2 The elliptic modulus and exact nome inverse

Let

$$m = k^2 = \frac{\vartheta_2(0, q)^4}{\vartheta_3(0, q)^4}.$$

For  $0 < m < 1$ , the nome is exactly

$$q = \exp\left(-\pi \frac{K(1-m)}{K(m)}\right), \quad (17.4)$$

where  $K$  is the complete elliptic integral. Its small- $m$  expansion is

$$q = \frac{m}{16} + 8\left(\frac{m}{16}\right)^2 + 84\left(\frac{m}{16}\right)^3 + 992\left(\frac{m}{16}\right)^4 + 12514\left(\frac{m}{16}\right)^5 + \cdots. \quad (17.5)$$

This is an instructive case where a difficult-looking inverse  $q$ -series has an exact representation in a different special-function language.

### 17.3 The modular endpoint $q \rightarrow 1$

Set  $q = e^{-\pi t}$ . Modular transformation gives

$$\vartheta_3(0, e^{-\pi t}) = t^{-1/2} \vartheta_3(0, e^{-\pi/t}) = t^{-1/2} \left(1 + 2e^{-\pi/t} + \dots\right).$$

Thus for a large target  $y$ ,

$$t = y^{-2} \left(1 + O(e^{-\pi y^2})\right), \quad q = \exp(-\pi y^{-2} + O(e^{-\pi y^2})).$$

The inverse correction is exponentially small, not algebraic. Similar modular formulas treat  $\vartheta_2$  and  $\vartheta_4$ , with branch changes under the modular group.

### 17.4 Root-of-unity and modular-sheet structure

Theta functions possess both product expansions and modular transformations. Near a rational point of the modular parameter, the most stable inverse is usually obtained by applying a modular transformation that sends the cusp to infinity, using a small-nome inverse there, and transforming back. The modular matrix is part of the inverse branch label. This is the theta-function analog of recording the primitive root  $\zeta$  and radial direction for a  $q$ -Pochhammer inverse.



## **Part V**

# **The Fabius–Rvachev deformation and inverse parameter recovery**

## Chapter 18

# A geometrically weighted atomic-function family

### 18.1 Definition by a random series

Let  $U_0, U_1, \dots$  be independent uniform random variables on  $[-1, 1]$ , and let  $0 < q < 1$ . Define

$$X_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (18.1)$$

The weights are positive and sum to one, so  $X_q \in [-1, 1]$ . Its characteristic function is

$$\widehat{u}_q(t) = \mathbb{E} e^{itX_q} = \prod_{j=0}^{\infty} \text{sinc}((1 - q)q^j t), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (18.2)$$

The product decays faster than every negative power on the real axis, and therefore  $X_q$  has a compactly supported  $C^\infty$  density  $u_q$ . The distributional identity

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q$$

with an independent copy  $X'_q$  yields the refinement equation

$$\widehat{u}_q(t) = \text{sinc}((1 - q)t) \widehat{u}_q(qt). \quad (18.3)$$

At  $q = 1/2$ , the weights are  $2^{-j-1}$ , and  $u_{1/2}$  is the standard Rvachev up-density on  $[-1, 1]$ . If

$$F_q(x) = \Pr(X_q \leq 2x - 1), \quad 0 \leq x \leq 1,$$

then  $F_{1/2}$  is the Fabius function. On the left half of the support,  $u_{1/2}(t) = F_{1/2}(t + 1)$ ; symmetry supplies the right half.

### 18.2 Exact cumulants

Cumulants add under independent sums and scale homogeneously. For  $U \sim \text{Unif}[-1, 1]$ ,

$$\kappa_{2m}(U) = \frac{2^{2m} B_{2m}}{2m}, \quad \kappa_{2m+1}(U) = 0. \quad (18.4)$$

Therefore

$$\kappa_{2m}(X_q) = \kappa_{2m}(U) \frac{(1-q)^{2m}}{1-q^{2m}}, \quad \kappa_{2m+1}(X_q) = 0. \quad (18.5)$$

The factors  $(1-q)^{2m}/(1-q^{2m})$  are elementary  $q$ -analogs, and their parameter inverses are part of the same theory developed earlier.

**Theorem 18.1** (PROVED Exact variance inverse). *The variance and its inverse are*

$$V(q) = \text{Var } X_q = \frac{1-q}{3(1+q)}, \quad q = \frac{1-3V}{1+3V}. \quad (18.6)$$

*In particular, the Rvachev value  $q = 1/2$  is characterized by  $V = 1/9$ .*

*Proof.* Use  $\text{Var } U = 1/3$  and sum the geometric series of squared weights:

$$\text{Var } X_q = \frac{1}{3}(1-q)^2 \sum_{j \geq 0} q^{2j} = \frac{1-q}{3(1+q)}.$$

Solving the resulting linear-fractional equation gives the inverse. □

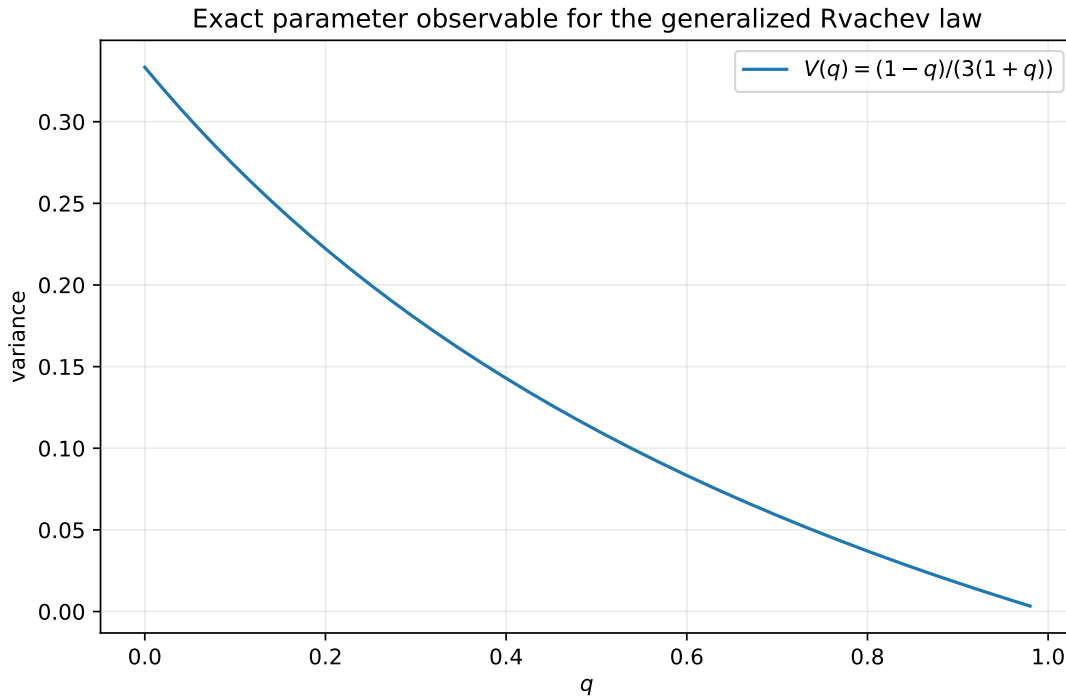


Figure 18.1: The exact variance observable for the geometric atomic-function family. It maps  $q \in (0, 1)$  monotonically onto  $(0, 1/3)$ .

### 18.3 Every even cumulant is an inverse observable

Normalize

$$R_m(q) = \frac{\kappa_{2m}(X_q)}{\kappa_{2m}(U)} = \frac{(1-q)^{2m}}{1-q^{2m}}.$$

**Theorem 18.2** (PROVED Monotone cumulant recovery). *For every  $m \geq 1$ ,  $R_m$  is strictly decreasing from one to zero on  $(0, 1)$ . Hence any nonzero even cumulant determines  $q$  uniquely through*

$$R_m(1 - q^{2m}) = (1 - q)^{2m}. \quad (18.7)$$

*The endpoint inverse asymptotics are*

$$q = \frac{1 - R_m}{2m} + O((1 - R_m)^2), \quad R_m \uparrow 1, \quad (18.8)$$

$$1 - q = (2mR_m)^{1/(2m-1)} \left(1 + O(R_m^{1/(2m-1)})\right), \quad R_m \downarrow 0. \quad (18.9)$$

*Proof.* Logarithmic differentiation gives

$$\frac{R'_m}{R_m} = -\frac{2m}{1 - q} + \frac{2mq^{2m-1}}{1 - q^{2m}} < 0,$$

because  $q^{2m-1}(1 - q) < 1 - q^{2m}$ . The endpoint expansions follow by ordinary and Puiseux reversion.  $\square$

## 18.4 Scale-free recovery from kurtosis

Suppose the observed variable is  $Y = cX_q$  with unknown nonzero scale  $c$ . Variance alone no longer identifies  $q$ , but standardized cumulants remove  $c$ . The excess kurtosis is

$$\gamma_2(q) = \frac{\kappa_4(X_q)}{\kappa_2(X_q)^2} = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}. \quad (18.10)$$

It has the exact inverse

$$q = \sqrt{\frac{1 + 5\gamma_2/6}{1 - 5\gamma_2/6}}, \quad -\frac{6}{5} < \gamma_2 < 0. \quad (18.11)$$

The standardized sixth cumulant supplies an independent check:

$$\frac{\kappa_6(X_q)}{\kappa_2(X_q)^3} = \frac{48}{7} \frac{(1 - q^2)^2}{1 + q^2 + q^4}. \quad (18.12)$$

Thus the deformation parameter remains exactly identifiable even when the support scale is unknown.

## 18.5 Conditioning

From (18.6),

$$\frac{dq}{dV} = -\frac{6}{(1 + 3V)^2}.$$

Absolute recovery remains regular throughout  $0 < V < 1/3$ . The relative condition number diverges as  $q \downarrow 0$ , because relative error in a parameter tending to zero is intrinsically severe; this is not an absolute instability.

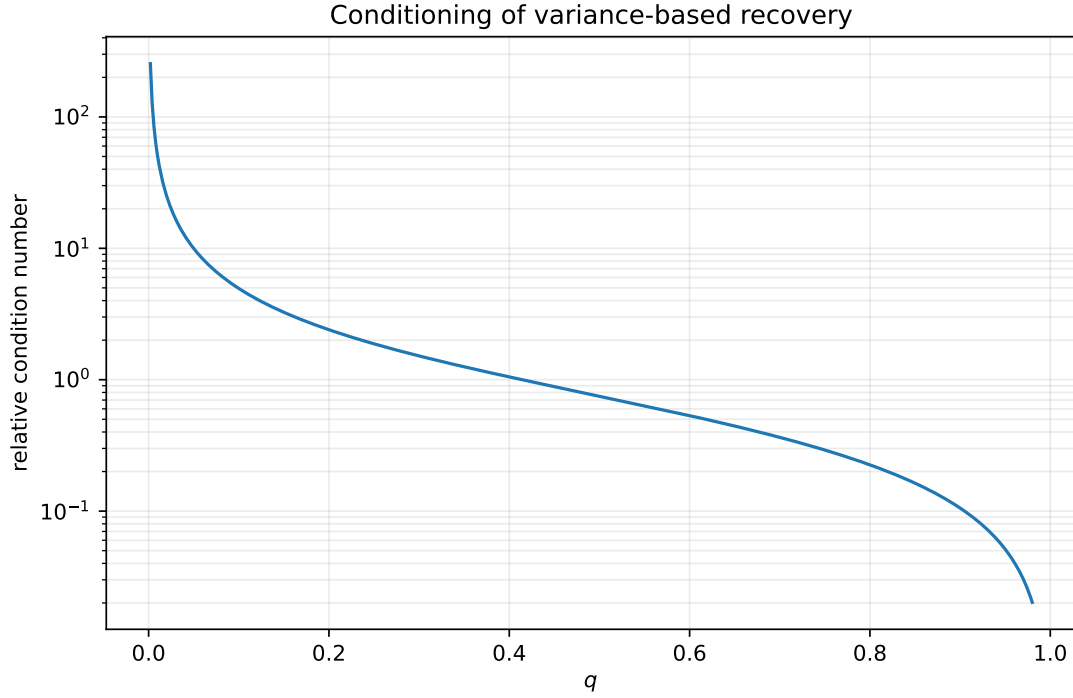


Figure 18.2: Relative condition number of variance-based recovery. The divergence at  $q = 0$  is caused by normalization by the vanishing recovered parameter.

## 18.6 Parameter derivatives of the characteristic function

Let

$$w_j(q) = (1 - q)q^j, \quad w'_j(q) = q^{j-1}(j - (j + 1)q).$$

Away from zeros of the sinc factors,

$$\partial_q \log \hat{u}_q(t) = \sum_{j \geq 0} w'_j(q) \left( t \cot(w_j(q)t) - \frac{1}{w_j(q)} \right). \quad (18.13)$$

This is a rapidly convergent even-power series near  $t = 0$ , and its coefficients are derivatives of (18.5). Fourier inversion then gives the tangent density  $\partial_q u_q$  and tangent CDF  $\partial_q F_q$ .

If  $x_p(q)$  is a quantile defined by

$$\Pr(X_q \leq x_p(q)) = p,$$

then, wherever  $u_q(x_p) > 0$ ,

$$x'_p(q) = -\frac{\partial_q \Pr(X_q \leq x)|_{x=x_p}}{u_q(x_p)}. \quad (18.14)$$

This is the ordinary partial-inverse sensitivity formula in a probabilistic disguise.

## 18.7 Limits $q \rightarrow 0$ and $q \rightarrow 1$

As  $q \downarrow 0$ ,  $X_q \rightarrow U_0$ , so the density approaches the uniform density on  $[-1, 1]$ . Corrections can be developed by expanding the refinement equation in powers of  $q$ , with boundary layers at the support endpoints.

As  $q \uparrow 1$ ,  $X_q \rightarrow 0$  in probability because its variance tends to zero. After standardization,

$$Z_q = \frac{X_q}{\sqrt{V(q)}},$$

the largest normalized summand tends to zero, and the triangular-array central limit theorem gives  $Z_q \Rightarrow N(0, 1)$ . The standardized cumulants in (18.5) quantify Edgeworth corrections and can themselves be inverted to estimate  $q$  from a nearly Gaussian observed profile.

# Chapter 19

## Inverse Fabius and Rvachev observables

### 19.1 Recovering $q$ from moments

At the exact Rvachev member,  $q = 1/2$ , the moment and cumulant identities give an overdetermined set of checks:

$$\text{Var } X_{1/2} = \frac{1}{9}, \quad \gamma_2(X_{1/2}) = -\frac{18}{25},$$

with the sixth standardized cumulant obtained from (18.12). If numerically computed Fabius or up values are used to estimate moments, disagreement among these recovered values of  $q$  measures quadrature or truncation error.

### 19.2 Recovery from a CDF value

Fix  $x \in (0, 1)$ , and consider

$$q \mapsto F_q(x) = \Pr(X_q \leq 2x - 1).$$

Unlike the variance map, this need not be globally monotone for every  $x$ . A local inverse around  $q_0$  exists when  $\partial_q F_q(x)|_{q_0} \neq 0$ , and its first terms are

$$\begin{aligned} q - q_0 &= \frac{p - F_{q_0}(x)}{\partial_q F_{q_0}(x)} \\ &\quad - \frac{\partial_q^2 F_{q_0}(x)}{2(\partial_q F_{q_0}(x))^3} (p - F_{q_0}(x))^2 + \dots \end{aligned} \quad (19.1)$$

The derivatives can be obtained from (18.13), from differentiated refinement equations, or by common-random-number Monte Carlo.

CDF inversion becomes ill-conditioned near symmetry points forced to be constant. For example,

$$F_q(1/2) = \Pr(X_q \leq 0) = \frac{1}{2}$$

for every  $q$ , so the midpoint carries no parameter information. This is a simple but important experimental-design constraint.

### 19.3 Recovery from several samples

Given values  $F_q(x_1), \dots, F_q(x_r)$ , one can solve a least-squares or maximum-likelihood problem in  $q$ . The local Fisher-like information for independent measurements with variances  $\sigma_i^2$  is

$$I(q) = \sum_{i=1}^r \frac{(\partial_q F_q(x_i))^2}{\sigma_i^2}.$$

Sampling points should be placed where the tangent CDF is large and linearly independent of nuisance directions such as translation and scale. Variance and kurtosis provide analytically orthogonalized alternatives.

### 19.4 A connection to geometric-comb interpolation

The weights  $(1 - q)q^j$  form a geometric comb. Every moment derivative, characteristic-function jet, and refinement iterate samples analytic functions on this comb. Newton interpolation on nodes  $q^j$  therefore provides a natural discrete representation of tangent observables. Its divided differences are  $q$ -differences, and the products

$$\prod_{j=0}^{n-1} (x - q^j) = x^n(x^{-1}; q)_n$$

are finite  $q$ -Pochhammer factors. Consequently, inversion of the Fabius–Rvachev deformation and inversion of  $q$ -products are linked at the level of both probability and interpolation: the same geometric node set controls cumulants, Newton bases, and parameter sensitivity.

### 19.5 Rigorous and conjectural geometry of CDF branches

**Conjecture 19.1** (**CONJECTURAL** Finite-fold CDF inverse). For each fixed  $x \in (0, 1) \setminus \{1/2\}$ , the map  $q \mapsto F_q(x)$  has only finitely many critical points on every compact subinterval of  $(0, 1)$ , all generically simple. At a simple critical point, the inverse has the universal square-root form. Exceptional higher singularities occur only at symmetry-enforced or refinement-resonant values of  $x$ .

**Conjecture 19.2** (**CONJECTURAL** Total cumulant identifiability). For the larger family

$$X_{q,c,\mu} = \mu + c \sum_{j \geq 0} q^j U_j,$$

the first moment, variance, and kurtosis form a globally injective coordinate system on  $\mu \in \mathbb{R}$ ,  $c > 0$ ,  $0 < q < 1$ . The explicit inverse consists of translation from the mean,  $q$  from (18.11), and  $c$  from the variance.

The last statement is in fact immediate at the algebraic level; the conjectural aspect concerns stable identifiability under natural broader atomic-digit perturbations, where the digit law is not known exactly.



## **Part VI**

# **Branch-aware computation, certification, and reproducibility**

## Chapter 20

# Stable numerical inversion

### 20.1 The forward map must be evaluated in inverse coordinates

An inverse algorithm is only as reliable as its forward evaluator. Directly multiplying many nearly unit factors, subtracting two nearly equal  $q$ -gamma logarithms, or forming a Gaussian coefficient as a quotient of three small products can destroy the digits needed by Newton correction. The following representation rules are therefore part of the inverse theory, not mere implementation details.

For a finite product away from its zeros, use

$$\log(a; q)_n = \sum_{j=0}^{n-1} \log(1 - aq^j), \quad (20.1)$$

with a branch-continuous complex logarithm and a  $\log 1p$ -type implementation. If a factor is close to zero, remove it explicitly:

$$(a; q)_n = (1 - aq^r) \prod_{\substack{0 \leq j < n \\ j \neq r}} (1 - aq^j).$$

This exposes the local zero and prevents a small factor from being hidden inside a large logarithmic sum.

For an infinite product with  $0 < q < 1$ , truncate only after bounding the tail. For  $0 \leq a < 1$ ,

$$0 \leq - \sum_{j=N}^{\infty} \log(1 - aq^j) \leq \frac{aq^N}{(1 - aq^N)(1 - q)}. \quad (20.2)$$

Indeed,  $-\log(1 - x) \leq x/(1 - x)$ , and the remaining geometric sum gives the right-hand side. The same argument applies to absolute values when  $|aq^N| < 1$ . Thus a requested forward tolerance can be converted into a certified truncation index before inversion begins.

For Gaussian coefficients, the product quotient

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j}$$

is stable in logarithmic form away from roots of unity. Near  $q = 1$ , use the analytic function

$$\text{exprel}(z) = \frac{e^z - 1}{z}$$

and write

$$\log \left[ \begin{matrix} n \\ k \end{matrix} \right]_q = \log \binom{n}{k} + \sum_{j=1}^k [\log \text{exprel}((n - k + j)t) - \log \text{exprel}(jt)], \quad q = e^t. \quad (20.3)$$

At a root of unity, cancel cyclotomic factors symbolically or group exponents by residue class before numerical evaluation.

## 20.2 Choice of local coordinate

The natural unknown is rarely  $q$  itself over the entire domain. The following coordinates regularize the main endpoints:

| regime                 | coordinate                   | reconstruction     | reason                                     |
|------------------------|------------------------------|--------------------|--------------------------------------------|
| $q \approx 0$          | $u = q^{1/d}$ if needed      | $q = u^d$          | resolves fractional Newton polygons        |
| $q \approx 1$          | $t = \log q$ or $-\log q$    | $q = e^t$          | Bernoulli expansions are regular in $t$    |
| $q \approx -1$         | $t = q + 1$ or $q = -e^{-t}$ | direct or radial   | separates algebraic and radial limits      |
| $q \approx \zeta$      | $t = -\log(q/\zeta)$         | $q = \zeta e^{-t}$ | fixes a cyclotomic ray and logarithm sheet |
| $q \rightarrow \infty$ | $r = q^{-1}$                 | $q = r^{-1}$       | turns infinity into an ordinary origin     |
| positive $q$ globally  | $t = \log q$                 | $q = e^t$          | preserves positivity and reciprocity       |

If the first nonzero forward term is  $cu^m$ , Newton should be applied in  $v = u^m$ , or the sheet should be initialized by the corresponding Puiseux root. Applying ordinary Newton directly in  $u$  at the critical point loses quadratic convergence because the derivative vanishes.

## 20.3 Logarithmic Newton and Halley corrections

Suppose a nonzero target  $y$  is to be matched by a product-like function  $F(p)$ . On a chosen logarithm sheet define

$$h(p) = \text{Log } F(p) - \text{Log } y.$$

Then logarithmic Newton is

$$p_{r+1} = p_r - \frac{h(p_r)}{h'(p_r)} = p_r - \frac{\text{Log}(F(p_r)/y)}{F'(p_r)/F(p_r)}. \quad (20.4)$$

It avoids overflow and measures relative rather than absolute error. For the finite Pochhammer base inverse,

$$\partial_q \log(a; q)_n = -a \sum_{j=1}^{n-1} \frac{jq^{j-1}}{1 - aq^j}, \quad (20.5)$$

while for argument inversion,

$$\partial_a \log(a; q)_n = - \sum_{j=0}^{n-1} \frac{q^j}{1 - aq^j}. \quad (20.6)$$

These expressions should be evaluated together with the logarithm so that the same powers and factors are reused.

Where second derivatives are inexpensive, Halley's correction

$$p_{r+1} = p_r - \frac{2hh'}{2(h')^2 - hh''} \quad (20.7)$$

can be effective after a branch-safe Newton iterate has entered the local basin. It is not a substitute for bracketing or continuation: close to a critical value, the denominator in (20.7) may itself be small and a cubic method can jump farther than Newton.

## 20.4 Predictor–corrector continuation of a selected sheet

Let a branch be defined by

$$F(p(s), \lambda(s)) = y(s).$$

As long as  $F_p \neq 0$ , differentiation gives the continuation equation

$$p'(s) = \frac{y'(s) - F_\lambda(p(s), \lambda(s))\lambda'(s)}{F_p(p(s), \lambda(s))}. \quad (20.8)$$

For several auxiliary parameters, replace the numerator by  $y' - \sum_j F_{\lambda_j} \lambda'_j$ . A practical branch tracker uses one Runge–Kutta or Taylor predictor for (20.8), followed by a Newton corrector at the new value of  $s$ . The step size is reduced when  $|F_p|$  decreases or the Newton correction becomes a substantial fraction of the predicted displacement.

A logarithm or argument used in the forward function must be unwrapped continuously along the path. Re-evaluating every logarithm on its principal sheet at every step changes the analytic function being followed and can silently transport the iterate to a different inverse branch.

## 20.5 Continuation through a fold

At a simple fold  $(p_*, y_*)$ , introduce a local square coordinate

$$y - y_* = \sigma^2, \quad p - p_* = c_1\sigma + c_2\sigma^2 + \cdots.$$

Continuation in  $\sigma$ , rather than  $y$ , makes the selected sheet regular. Alternatively, use pseudo-arclength continuation for the curve  $F(p) - y = 0$ : append a linear arclength condition and solve for  $(p, y)$  together. The latter passes turning points without choosing a square-root sheet in advance.

At a cubic or higher point, the appropriate coordinate is obtained from the Newton polygon. The exact identity

$$(-1/3; q)_3 = \frac{32}{27} + \frac{4}{27}(q+1)^3$$

provides an ideal regression test: a correct branch tracker should recover all three cube-root sheets and their cyclic monodromy around  $y = 32/27$ .

## 20.6 Certified real inverses

For a continuous strictly monotone real branch, interval bisection is the fallback method. Asymptotics supply a short bracket and Newton supplies speed, but monotonicity supplies the proof.

**Theorem 20.1** (PROVED Residual-to-root enclosure). *Let  $f$  be differentiable on an interval  $I$ , suppose  $f(p) = y$  has a solution  $p \in I$ , and assume*

$$\inf_{x \in I} |f'(x)| \geq m > 0.$$

*Then every approximation  $p_0 \in I$  satisfies*

$$|p - p_0| \leq \frac{|f(p_0) - y|}{m}. \quad (20.9)$$

*Proof.* The mean-value theorem gives  $f(p_0) - f(p) = f'(\xi)(p_0 - p)$  for some  $\xi$  between  $p$  and  $p_0$ . Taking absolute values proves the claim.  $\square$

For logarithmic inversion, apply the theorem to  $\log f$  and obtain a bound from the relative residual. Near a fold,  $m$  necessarily tends to zero; the square coordinate restores a nonvanishing derivative and a meaningful bound.

Interval Newton sharpens bisection. Given an interval  $X$ , midpoint  $x_0$ , and an interval extension  $f'(X)$ , set

$$N(X) = x_0 - \frac{f(x_0) - y}{f'(X)}.$$

If  $N(X) \subseteq X$ , a root is enclosed; if  $N(X) \subset \text{int } X$  and zero is excluded from  $f'(X)$ , the root is unique in  $X$ . Directed rounding is essential when the result is presented as a proof rather than as a high-precision observation.

## 20.7 A complex disk certificate

The following elementary Rouché test certifies one local inverse sheet.

**Theorem 20.2** (PROVED One-zero disk). *Let  $F$  be holomorphic on the closed disk  $|p - p_0| \leq r$ . Suppose*

$$\sup_{|p-p_0| \leq r} |F''(p)| \leq M$$

and

$$|F(p_0) - y| + \frac{1}{2}Mr^2 < |F'(p_0)|r. \quad (20.10)$$

*Then  $F(p) - y$  has exactly one zero, counted with multiplicity, in the disk.*

*Proof.* Taylor's formula with an integral remainder bounds

$$|F(p) - F(p_0) - F'(p_0)(p - p_0)| \leq \frac{1}{2}Mr^2$$

on the boundary. By (20.10), the difference between  $F(p) - y$  and  $F'(p_0)(p - p_0)$  is smaller than the latter. Rouché's theorem gives the same number of zeros, namely one.  $\square$

This test is especially convenient after a truncated inverse series has provided  $p_0$ . Bounds for  $F''$  follow directly from the product's logarithmic derivatives or from Cauchy estimates on a slightly larger disk.

## 20.8 Discriminants, resultants, and a zero-fiber factorization

For finite products, algebraic singularities can be found before numerical continuation. The critical-value polynomial is

$$\Delta_n(q, y) = \text{Res}_a((a; q)_n - y, \partial_a(a; q)_n).$$

Its zeros are precisely the finite branch values, with multiplicities. A subresultant sequence is usually preferable to expanding the full determinant.

At  $y = 0$ , the discriminant has a particularly transparent factorization.

**Theorem 20.3** (PROVED Zero-fiber argument discriminant). *For  $n \geq 2$ , viewed as a polynomial in  $a$ ,*

$$\text{Disc}_a(a; q)_n = q^{n(n-1)(n-2)/3} \prod_{d=1}^{n-1} (1 - q^d)^{2(n-d)}. \quad (20.11)$$

*Proof.* The leading coefficient is  $(-1)^n q^{n(n-1)/2}$ , and the roots are  $q^{-j}$ ,  $0 \leq j < n$ . Hence

$$\text{Disc}_a(a; q)_n = q^{n(n-1)^2} \prod_{0 \leq i < j < n} (q^{-i} - q^{-j})^2.$$

Writing  $q^{-i} - q^{-j} = q^{-j}(q^{j-i} - 1)$ , collecting the powers of  $q$ , and noting that the difference  $d = j - i$  occurs  $n - d$  times gives (20.11).  $\square$

The theorem simultaneously records every collision of argument zeros at roots of unity. It also gives the exact vanishing order of the discriminant, which is useful for deciding how many precision bits are lost near a cyclotomic collision.

## 20.9 Certification of discrete parameter recovery

A continuous interpolation in the order or in the Gaussian parameters is a branch-finding device; the original problem may still require an integer. Suppose an interpolated inverse gives  $x_*$  with an error bound  $\eta < 1/2$ . There is then at most one candidate integer  $m$  with  $|m - x_*| \leq \eta$ . Evaluate the exact recurrence at that candidate to certify it.

For finite products,

$$\frac{(a; q)_{n+1}}{(a; q)_n} = 1 - aq^n.$$

For Gaussian coefficients,

$$\frac{\begin{bmatrix} n+1 \\ k \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1 - q^{n+1}}{1 - q^{n+1-k}}, \quad \frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}_q}{\begin{bmatrix} n \\ k \end{bmatrix}_q} = \frac{1 - q^{n-k}}{1 - q^{k+1}}. \quad (20.12)$$

These ratios permit exact or interval comparisons without constructing large coefficients. If the target is noisy rather than exact, they also delimit the Voronoi interval in target space assigned to each integer parameter.

# Chapter 21

## Reproducible symbolic and numerical experiments

### 21.1 Scope and arithmetic model

The accompanying program `inverse_q_analogs_experiments.py` uses SymPy for exact rational and polynomial identities, mpmath at eighty decimal digits for transcendental evaluations, and NumPy/Matplotlib for tabulation and figures. It deliberately separates three kinds of evidence:

- (i) exact symbolic identities, simplified over  $\mathbb{Q}$  or  $\mathbb{Q}[a, q]$ ;
- (ii) asymptotic residuals computed at increasing precision and decreasing endpoint parameter;
- (iii) plots intended to reveal geometry, but never used as proofs.

Every plotted data set is also written as CSV, so the numerical content can be checked without trusting the graphics backend.

### 21.2 The exact $q = -1$ table

The script differentiates  $(a; q)_n$  symbolically, substitutes the critical parameter from the theorem in the finite-product chapter, and determines the first nonzero  $q$ -derivative. The resulting exact table is:

Table 21.1: Finite Pochhammer critical slices at  $q = -1$ .

| $n$ | $a_*$   | critical order | $(a_*; -1)_n$               |
|-----|---------|----------------|-----------------------------|
| 3   | $-1/3$  | 3              | $32/27$                     |
| 4   | $1/3$   | 2              | $64/81$                     |
| 5   | $-1/5$  | 2              | $3456/3125$                 |
| 6   | $1/5$   | 2              | $13824/15625$               |
| 7   | $-1/7$  | 2              | $884736/823543$             |
| 8   | $1/7$   | 2              | $5308416/5764801$           |
| 9   | $-1/9$  | 2              | $409600000/387420489$       |
| 10  | $1/9$   | 2              | $3276800000/3486784401$     |
| 11  | $-1/11$ | 2              | $298598400000/285311670611$ |

The first row detects the exact cubic at  $n = 3$ ; every subsequent displayed row has critical order two. The alternating location of  $a_*$ , together with the limiting approach  $a_* \rightarrow 0$ , is shown in [Figure 21.1](#).

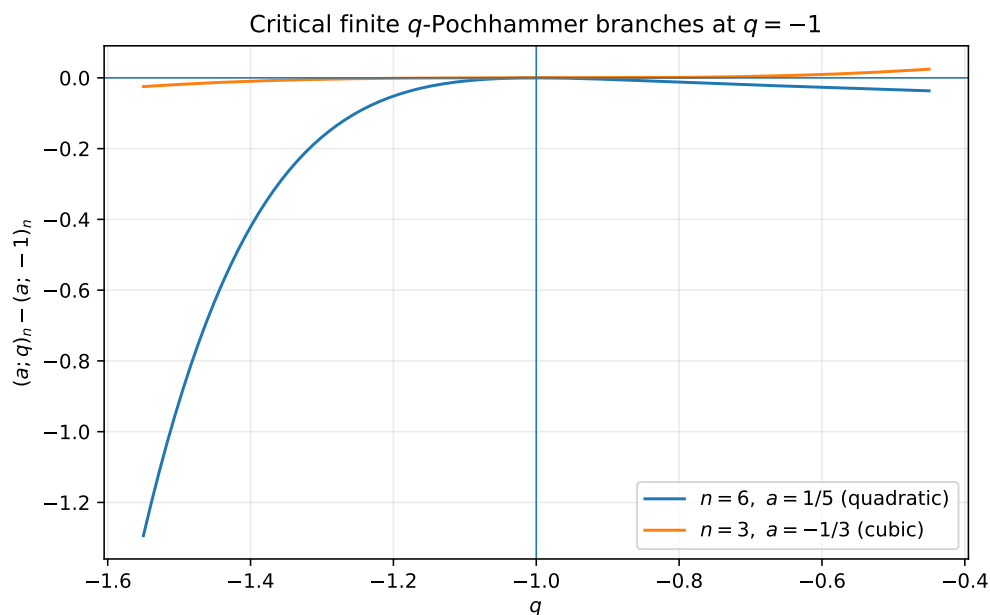


Figure 21.1: The distinguished critical argument at  $q = -1$  for finite Pochhammer products. Even and odd orders approach the origin from opposite sides.

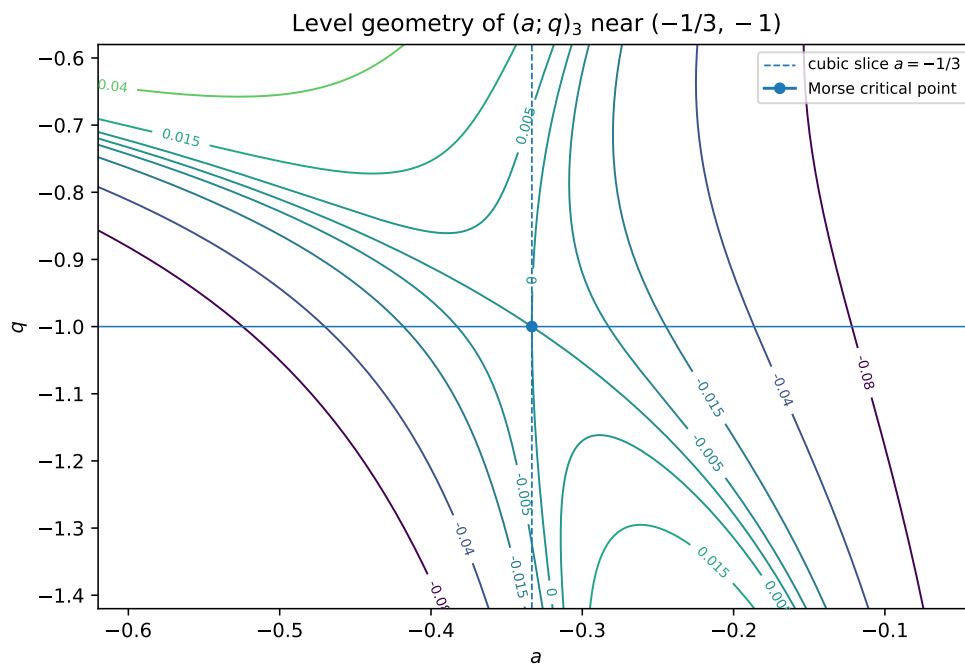


Figure 21.2: Local level geometry of  $(a; q)_3$  near  $(a, q) = (-1/3, -1)$ . The fixed- $a$  slice is cubic, although the full critical point is a nondegenerate saddle.



### 21.3 Endpoint inverse residuals

For the real rays  $q = e^{-t}$  and  $q = -e^{-t}$ , the program evaluates a long truncated product with a tail bound, forms the target logarithm, and compares both the leading and corrected inverse formulas. At  $a = 0.37$  and  $t = 0.012$ , the corrected  $q \rightarrow 1$  inverse has relative error approximately  $4.45 \times 10^{-10}$ , while the leading term has error approximately  $1.71 \times 10^{-5}$ . The decay over the complete test grid is displayed in Figure 21.3.

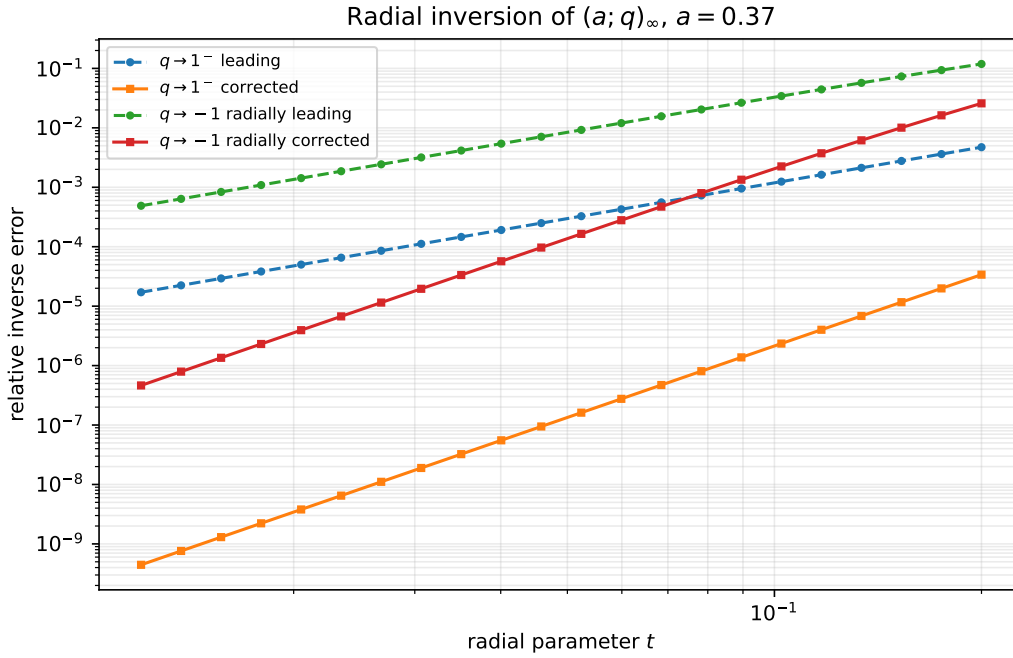


Figure 21.3: Relative errors of leading and corrected radial inverse formulas. The slopes diagnose the first omitted asymptotic order.

The primitive-root experiment uses  $m = 3, 4, 5$ . After subtracting the  $t^{-1}$  term and the explicit constant  $C_\zeta(a)$ , the residual divided by  $t$  approaches a finite complex limit, in agreement with the predicted coefficient  $-\mathcal{D}_\zeta(a)$ . This test is sensitive to the sign in the Euler–Maclaurin coefficient and to the chosen logarithm sheet.

### 21.4 The two small- $q$ regimes of $\Gamma_q$

Near  $q = 0$ ,  $q^x$  and  $q$  compete. The experiment samples  $x < 1$ ,  $x = 1$ ,  $1 < x < 2$ ,  $x = 2$ , and  $x > 2$ , and compares the exact logarithm with the Newton-polygon model. The distinct power laws are visible in Figure 21.4.

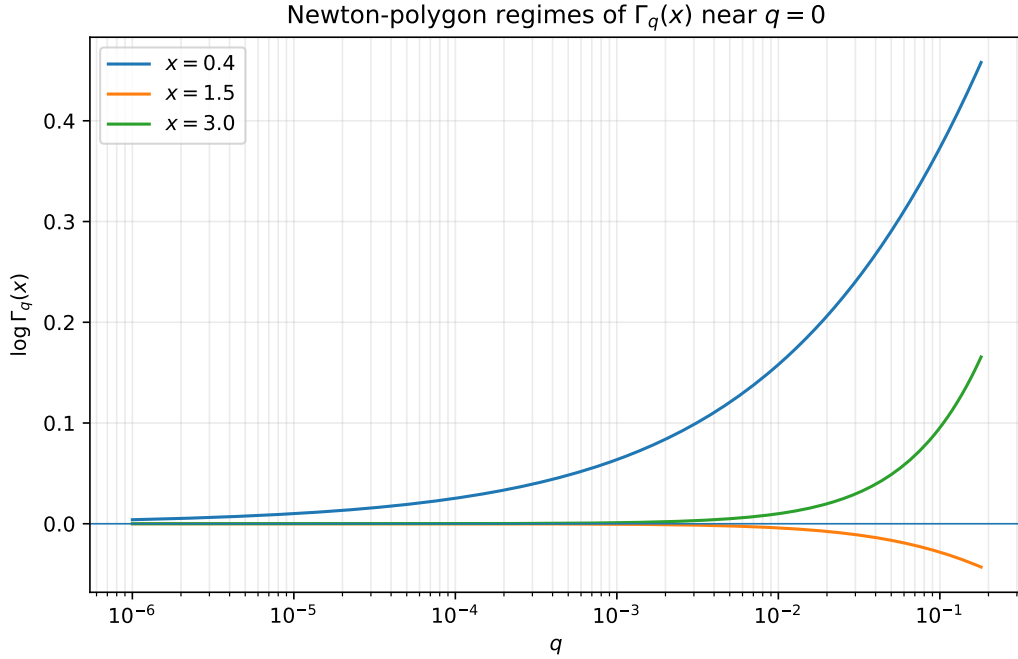


Figure 21.4: Residual scaling for  $\log \Gamma_q(x)$  as  $q \rightarrow 0$ . The competition between  $q^x$  and integer powers of  $q$  changes at integer values of  $x$ .

Near  $q = 1$ , the second-order model

$$\log \frac{\Gamma_{e^{-t}}(x)}{\Gamma(x)} = -\frac{(x-1)(x-2)}{4}t + \frac{(x-1)(x-2)(2x+3)}{144}t^2 + O(t^3)$$

was tested at  $x = 1/2, 3/2, 3, 17/4$ . The residual is smaller than  $3 \times 10^{-7}$  even at  $t = 0.05$  on this grid, and decreases at least cubically over the sampled range. The exact identities  $\Gamma_q(1) = \Gamma_q(2) = 1$  explain why those two values require separate inverse normal forms.

## 21.5 Gaussian inverse jets

The program tests five distinct base regimes for  $\left[\begin{smallmatrix} 8 \\ 3 \end{smallmatrix}\right]_q$ : ordinary reversion at zero, Bernoulli reversion at one, the nonzero and zero branches at minus one, and reciprocal reversion at infinity. Representative errors are shown in Table 21.2. Their purpose is not to optimize each truncation order, but to exercise all branch initializers through one common forward evaluator.

Table 21.2: Representative truncated inverse-jet errors for  $\left[\begin{smallmatrix} 8 \\ 3 \end{smallmatrix}\right]_q$ .

| regime                   | true $q$ | absolute error                       |
|--------------------------|----------|--------------------------------------|
| $q = 0$                  | 0.015    | $7.644\,590\,615\,83 \times 10^{-7}$ |
| $q = 1$                  | 1.012    | $3.651\,537\,610\,21 \times 10^{-8}$ |
| $q = -1$ , nonzero value | -0.992   | $2.454\,153\,191\,51 \times 10^{-4}$ |
| $q = -1$ , simple zero   | -0.992   | $4.330\,795\,781\,91 \times 10^{-4}$ |
| $q = \infty$             | 18       | $3.286\,441\,773\,85 \times 10^{-4}$ |

A stronger validation strategy is round-trip coefficient testing. If  $q = Q_N(y)$  is a truncated inverse series, substitute it symbolically into the truncated forward series and verify

$$F(Q_N(y)) - y = O((y - y_0)^{N+1}).$$

This catches algebraic errors that can be masked by a favorable numerical test point.

## 21.6 Exact recovery in the atomic-function family

The moment experiment evaluates

$$V(q) = \frac{1 - q}{3(1 + q)}$$

and the fourth and sixth cumulants at five values from  $q = 0.1$  to  $q = 0.93$ . Variance inversion returns the input to the full working precision. The map and its condition number are shown in [Figures 21.5 and 21.6](#).

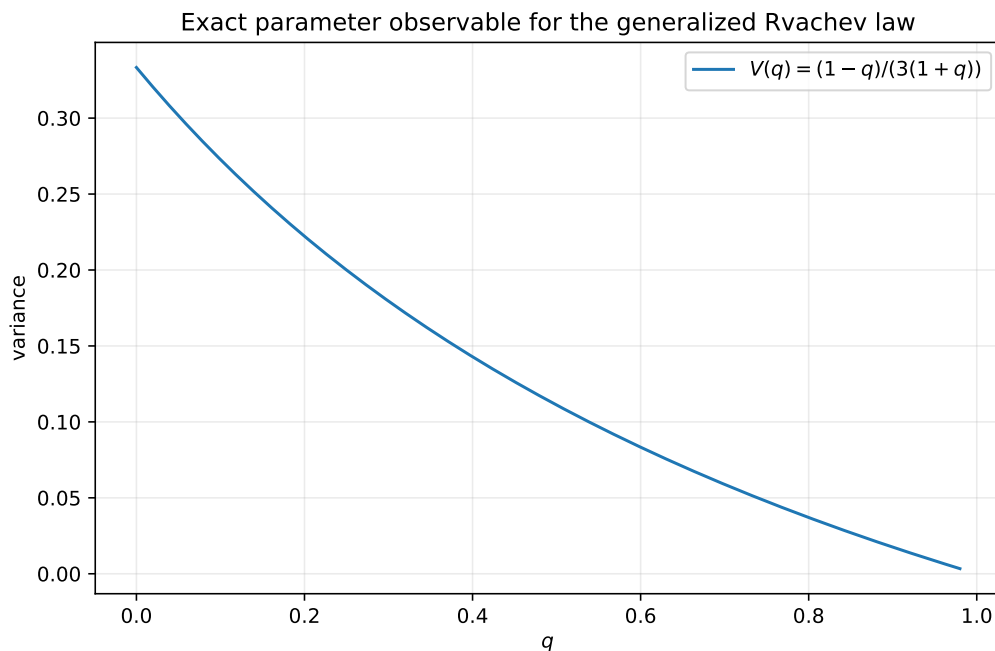


Figure 21.5: Exact inverse relation between the variance of the normalized geometric atomic family and its contraction ratio  $q$ .

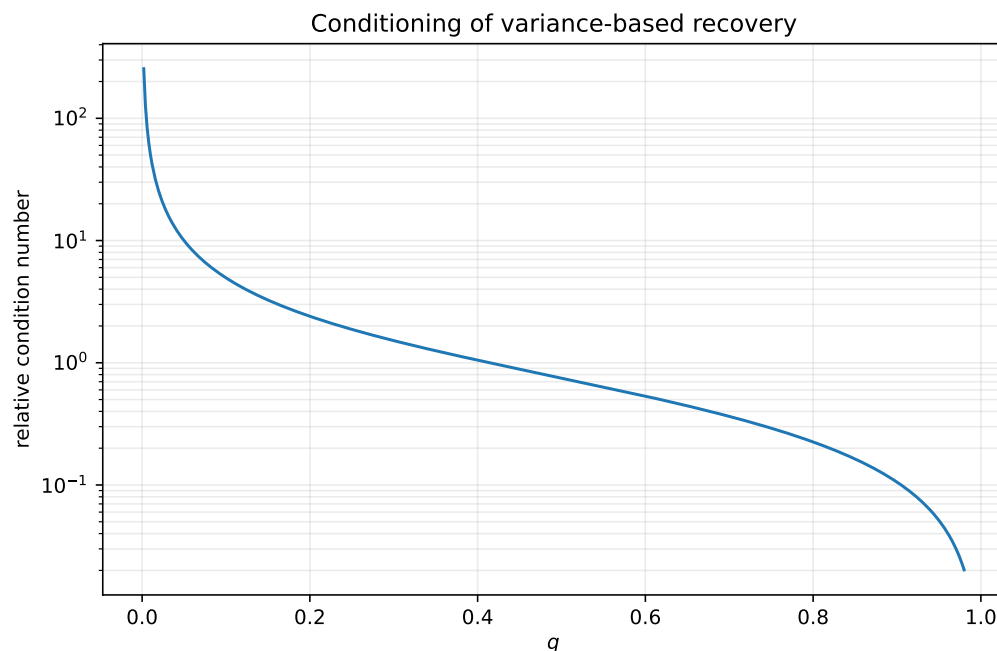


Figure 21.6: Conditioning of variance-based recovery. Absolute recovery becomes more sensitive as  $q \rightarrow 1$ , where the variance collapses.

Kurtosis inversion remains scale-free and therefore provides a genuinely independent check. In data analysis, comparing the variance-based and kurtosis-based values is more informative than reporting additional digits from either one alone.

## 21.7 Archive layout and reproduction

From the report directory, the numerical artifacts are regenerated with

```
1 python inverse_q_analogs_experiments.py
```

The script creates `data/*.csv`, both PDF and PNG versions of every figure, `numerical_results.tex`, and `numerical_summary.txt`. The only nonstandard dependencies are listed in `requirements.txt`. The LaTeX report then compiles without requiring Python; all generated files are included in the archive.

# Chapter 22

## An inversion atlas

### 22.1 Local normal forms by object and parameter

Table 22.1 summarizes the first coordinate to try. A formula in this table is a branch initializer; global uniqueness still requires the monotonicity or continuation analysis in the relevant chapter.

Table 22.1: Atlas of principal inverse normal forms.

| object                                   | recovered | point             | inverse coordinate or leading law                                                 |
|------------------------------------------|-----------|-------------------|-----------------------------------------------------------------------------------|
| $(a; q)_n$                               | $a$       | $a = 0$           | ordinary series in $1 - y$ , first term $(1 - y)/[n]_q$                           |
| $(a; q)_n$                               | $a$       | simple zero       | ordinary series in $y$ ; derivative is $-q^r \prod_{j \neq r} (1 - q^{j-r})$      |
| $(a; q)_n$                               | $a$       | root collision    | Puiseux power $y^{1/s}$ , where $s$ is the residue-class multiplicity             |
| $(a; q)_n$                               | $q$       | $q = 0$           | ordinary or Newton–Puiseux series selected by the least active exponent           |
| $(a; q)_n$                               | $q$       | $q = 1$           | ordinary series in $\log(y/(1 - a)^n)$ , unless the first cumulant vanishes       |
| $(a; q)_n$                               | $q$       | $q = -1$          | fold at $a_*(n)$ ; exact cubic only for $n = 3$ , $a = -1/3$                      |
| $(a; q)_\infty$                          | $a$       | $a = 0$           | ordinary series in $1 - y$                                                        |
| $(a; q)_\infty$                          | $a$       | $a = 1$           | ordinary series in $y/(q; q)_\infty$                                              |
| $(a; q)_\infty$                          | $q$       | $\zeta e^{-t}$    | reciprocal-log series $t \sim \text{Li}_2(a^m)/(m^2 X)$                           |
| $\begin{bmatrix} n \\ k \end{bmatrix}_q$ | $q$       | $q = 0$           | $q = (y - 1) + O((y - 1)^2)$                                                      |
| $\begin{bmatrix} n \\ k \end{bmatrix}_q$ | $q$       | $q = 1$           | $\log q \sim 2 \log(y/\binom{n}{k})/[k(n - k)]$                                   |
| $\begin{bmatrix} n \\ k \end{bmatrix}_q$ | $q$       | $q = \infty$      | reciprocal of the $q = 0$ inverse of $y/q^{k(n-k)}$                               |
| continuous Gaussian                      | upper     | finite point      | inverse $q$ -digamma difference; strictly increasing on the real principal domain |
| continuous Gaussian                      | lower     | center            | square-root fold about $w = z/2$                                                  |
| $[x]_q$                                  | $x$       | regular           | exact logarithmic formula $x = \log(1 - y + yq)/\log q$                           |
| $[x]_q$                                  | $q$       | $q = 1$           | ordinary series in $\log(y/x)$                                                    |
| $\Gamma_q(x)$                            | $x$       | minimum           | two square-root branches                                                          |
| $\Gamma_q(x)$                            | $q$       | $q = 0$           | generalized powers generated by $q^x$ and $q$                                     |
| $\Gamma_q(x)$                            | $q$       | $q = 1$           | ordinary unless $x = 1, 2$ , where the map is constant                            |
| $B_q(x, z)$                              | $x, z$    | regular           | inverse of a strict $q$ -digamma difference                                       |
| $B_q(x, z)$                              | $q$       | $q = 1$           | ordinary except on the critical curve $xz = 1$                                    |
| Euler $q$ -exponential                   | argument  | any regular point | exact reduction to the inverse infinite Pochhammer function                       |

| object        | recovered | point           | inverse coordinate or leading law                                             |
|---------------|-----------|-----------------|-------------------------------------------------------------------------------|
| theta/nome    | $q$       | $q = 0$         | odd or even power series according to the theta constant used                 |
| elliptic nome | $q$       | $k = 0$         | exact modular inverse $q = \exp[-\pi K(\sqrt{1-k^2})/K(k)]$                   |
| atomic family | $q$       | any $0 < q < 1$ | exact rational inverse from variance or exact algebraic inverse from kurtosis |

## 22.2 Regular, critical, and nonidentifiable parameters

Every entry belongs to one of three qualitatively different classes.

**Regular.** The partial derivative in the recovered parameter is nonzero. The inverse is analytic and its coefficients follow the universal Bell recurrence.

**Critical but identifiable.** The derivative vanishes, but a higher jet or several observables distinguish branches. Newton–Puiseux series and pseudo-arclength continuation are the appropriate tools.

**Nonidentifiable.** The forward map is locally constant in the selected parameter or two parameters enter only through a lower-dimensional combination. No numerical method can create information absent from the data; an additional observable or normalization is required.

The exact identities  $\Gamma_q(1) = \Gamma_q(2) = 1$ ,  $F_q(1/2) = 1/2$  for the symmetric atomic CDF, and  $[1]_q = 1$  are canonical nonidentifiable examples.

## 22.3 A worked multivariate recovery example

Consider the normalized atomic family with an unknown location, scale, and contraction ratio,

$$Y = \mu + cX_q, \quad c > 0.$$

Let  $m$ ,  $v$ , and  $g_2$  be its mean, variance, and excess kurtosis. Translation and scale invariance give

$$\mu = m, \quad q = \sqrt{\frac{1 + 5g_2/6}{1 - 5g_2/6}}, \quad c = \sqrt{\frac{v}{V(q)}}.$$

Thus a seemingly nonlinear three-parameter inverse becomes triangular after choosing invariants adapted to the symmetry group. This is a general lesson: before applying a multivariate Newton method, quotient out translations, scalings, reciprocal symmetries, and parameter permutations analytically.

## 22.4 A worked cyclotomic recovery example

Let  $q = \zeta e^{-t}$ , where  $\zeta = e^{2\pi i/3}$ , and suppose  $a$  is known with  $|a| < 1$ . For a very small observed product  $y$ , define

$$A = \frac{\text{Li}_2(a^3)}{9}, \quad X = -\text{Log } y + C_\zeta(a).$$

The two-term initializer is

$$t_0 = \frac{A}{X} + \frac{\mathcal{D}_\zeta(a)A^2}{X^3}, \quad q_0 = \zeta e^{-t_0}.$$

One logarithmic Newton correction using the grouped residue-class product then produces a high-accuracy point on the chosen radial sheet. The same target  $y$  may have many nonradial preimages; the path  $q/\zeta \in (0, 1)$  is part of the inverse specification.

## **Part VII**

# **Open structure and a concrete research program**

## Chapter 23

# Algebraic and analytic frontiers

### 23.1 Generic monodromy of finite Pochhammer inverses

For fixed generic  $q$ , the polynomial equation

$$(a; q)_n = y$$

defines an  $n$ -sheeted cover of the target sphere. The critical points are solutions of a rational Stieltjes-type equation, but the critical values do not inherit the simple geometric ordering of the zeros. The natural global question is the monodromy group generated by loops around those values.

**Conjecture 23.1** (**CONJECTURAL** Generic full monodromy). For every  $n \geq 3$ , outside a proper algebraic exceptional set of bases  $q$ , the monodromy group of  $(a; q)_n - y = 0$  over the  $y$ -sphere is the full symmetric group  $S_n$ .

A feasible proof program is to show that all generic critical points are simple, so their local monodromies are transpositions, and then prove that the critical graph connecting the sheets is connected. Degeneracies at roots of unity and secondary critical-value collisions provide the exceptional set. Computations for moderate  $n$  should use exact specializations and certified Galois-group algorithms rather than floating-point root permutations alone.

### 23.2 Secondary discriminants and Maxwell loci

The ordinary discriminant records a repeated root of the inverse equation. A secondary discriminant records equality of two distinct critical values. If  $c_i(q)$  are the critical points of  $a \mapsto (a; q)_n$ , then the Maxwell condition is

$$(c_i(q); q)_n = (c_j(q); q)_n, \quad i \neq j.$$

Eliminating  $c_i, c_j$  gives an algebraic locus in  $q$ . Such a collision does not create a higher critical point of the forward map; instead, two branch points project to the same target value and the topology of natural branch cuts changes.

**Research problem 23.2.** Factor the secondary discriminant for  $(a; q)_n$  uniformly in  $n$ , identify its cyclotomic factors, and determine the number and asymptotic distribution of its positive roots in  $(0, 1)$ .

A promising approach is to express the critical equation in the elementary symmetric functions of the geometric roots and use subresultants twice: first to construct the critical-value polynomial, then to take its discriminant in  $y$ .



### 23.3 A double-scaling theory at colliding zeros

The radial root-of-unity expansion assumed fixed  $a$  with  $a^m \neq 1$ . Argument inversion at a cyclotomic collision assumes fixed  $q = \zeta$ . The missing transition regime lets both variables approach the singular set:

$$q = \zeta e^{-t}, \quad a = \zeta^{-r} e^{-\alpha t}.$$

Factors with index  $j \equiv r \pmod{m}$  become

$$1 - e^{-(\alpha+j)t},$$

and their product is governed by a scaled gamma function rather than by an ordinary Taylor coefficient.

**Conjecture 23.3** (**CONJECTURAL** Cyclotomic gamma transition). After extracting an explicit power of  $t$ , the double-scaled finite and infinite Pochhammer products admit uniform expansions whose leading factors are products of Euler gamma functions with fractional shifts determined by residue classes modulo  $m$ . Their parameter inverses interpolate uniformly between the fixed-argument reciprocal-log law and the fixed-base Puiseux law.

The conjecture can be attacked first for finite products by exact residue-class factorization, then for the infinite product by Euler–Maclaurin summation on each arithmetic progression.

### 23.4 Resurgence of radial inverse series

The perturbative expansion at  $q = \zeta e^{-t}$  is accompanied by terms exponentially small in  $1/t$ , arising from singularities of the summand after Poisson or Abel–Plana transformation. Formal inversion transports these terms into exponentially small corrections to  $t(y)$ . The nonlinear composition mixes instanton sectors but preserves their exponential scales.

**Research problem 23.4.** Construct the Borel transform of the radial logarithm, locate its first singularities for general  $a$  and  $m$ , and derive the induced Stokes jumps of the inverse base. Compare them with the finite cyclic quantum-dilogarithm factors appearing in root-of-unity limits [12, 13].

A numerically decisive test is hyperasymptotic: subtract the optimally truncated algebraic series and fit the remainder to the predicted exponential scale as  $t \downarrow 0$ .

### 23.5 Global phase diagrams for $q$ -gamma base inversion

The local coefficient at  $q = 1$ ,

$$-\frac{(x-1)(x-2)}{4},$$

changes sign at  $x = 1, 2$ , where the entire base dependence disappears. Near  $q = 0$ , the dominant term changes at  $x = 1$  because  $q^x$  overtakes or falls behind  $q$ . These transitions suggest a nontrivial global phase diagram for the number of real base solutions of

$$\Gamma_q(x) = y, \quad q \in (0, 1).$$

**Research problem 23.5.** For each fixed  $x > 0$ , determine the exact number of critical points of  $q \mapsto \Gamma_q(x)$  on  $(0, 1)$ , and classify all target values by the number of real inverse branches. Prove the expected monotonicity sectors or locate counterexamples with certified interval arithmetic.

The logarithmic derivative is a Lambert series, so complete-monotonicity and variation-diminishing methods are natural. The asymptotic work of Moak and McIntosh provides endpoint control [8, 10]; the remaining issue is global sign variation.

## 23.6 The critical curve of the $q$ -beta base inverse

The first  $q \rightarrow 1$  coefficient of  $\log(B_q(x, z)/B(x, z))$  vanishes on  $xz = 1$ . On this curve, the second coefficient determines a square-root or higher inverse depending on whether it also vanishes. Solving the simultaneous equations for the first several coefficients gives a finite algebraic stratification in the  $(x, z)$ -plane.

**Research problem 23.6.** Determine the complete degeneracy hierarchy of the  $q$ -beta base inverse at  $q = 1$ . In particular, decide whether any positive  $(x, z) \neq (1, 1)$  makes both the first and second nonconstant coefficients vanish, and classify the resulting Puiseux exponents.

Because the coefficients are Bernoulli polynomials, this problem is accessible to exact elimination before numerical exploration.

# Chapter 24

## Combinatorial and probabilistic frontiers

### 24.1 Discrete inverse Gaussian problems

For fixed positive  $q$ , the continuous interpolation  $\mathcal{B}_q(z, w)$  separates the analytic branch problem from the integer constraint. The discrete inverse asks for all integer pairs  $(n, k)$  with

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = y.$$

Symmetry forces  $(n, k) \leftrightarrow (n, n - k)$ , and special bases can create additional coincidences.

**Conjecture 24.1** (**CONJECTURAL** Generic discrete uniqueness). Fix a transcendental real  $q > 0, q \neq 1$ . Equality

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n' \\ k' \end{bmatrix}_q$$

for nontrivial indices implies  $(n', k') = (n, k)$  or  $(n, n - k)$ .

For transcendental  $q$ , equality is equality of Gaussian polynomials, so the problem reduces to whether two rectangular partition-generating polynomials can coincide. Degree, first coefficients, cyclotomic factorization, and the largest hook data provide successive invariants. This appears substantially more tractable than equality at one algebraic numerical base.

### 24.2 Inverse universality from coefficient distributions

A positive polynomial

$$P(q) = \sum_{j=0}^D c_j q^j$$

can be normalized at  $q = e^t$  as a moment-generating function for the random exponent  $J$  with probabilities proportional to  $c_j$ . Its inverse near  $q = 1$  depends on cumulants of  $J$ . If a sequence of  $q$ -analogs has centered, scaled exponent distributions converging to a Gaussian law, then the first inverse correction is universal and higher corrections are organized by Edgeworth cumulants.

**Research problem 24.2.** Prove uniform inverse limit theorems for Gaussian and multinomial coefficients when all rectangle or composition dimensions grow. Determine the largest target window in which a finite cumulant reversion remains uniform, and match it to large-deviation inversion outside that window.

The central window uses analytic cumulant bounds; the tails should be expressed through a Legendre transform and then inverted by saddle-point methods.

### 24.3 Statistical recovery of the Fabius–Rvachev contraction

Variance and kurtosis recover  $q$  exactly for the ideal geometric-uniform model, but numerical or empirical data introduce truncation, discretization, noise, and possible misspecification of the digit law. A robust inverse theory should recover  $q$  jointly with nuisance parameters and quantify model error.

**Research problem 24.3.** Derive the semiparametric information bound for estimating  $q$  when location and scale are unknown and the symmetric digit distribution belongs to a specified moment class. Compare moment, characteristic-function, and CDF estimators, especially near  $q = 1$ , where variance-based recovery is poorly conditioned.

The characteristic function offers a natural continuum of observables. Its score from (18.13) can be orthogonalized against location and scale scores, giving an efficient weighting rule in Fourier space.

### 24.4 Global geometry of CDF inverse branches

The refinement equation couples  $F_q(x)$  at geometrically rescaled arguments. Differentiating in  $q$  produces a linear inhomogeneous refinement equation for the tangent CDF. Critical points of  $q \mapsto F_q(x)$  are zeros of this tangent solution.

**Conjecture 24.4** (CONJECTURAL Tame tangent zero set). For algebraic  $x \in (0, 1) \setminus \{1/2\}$ , the tangent function  $q \mapsto \partial_q F_q(x)$  has finitely many zeros on every compact subinterval of  $(0, 1)$ , and no interior accumulation point.

One route is to establish real analyticity in  $q$  away from a controlled set of refinement resonances. Another is to approximate the refinement operator by finite transfer matrices with certified spectral error and study zeros of the resulting analytic determinants.

## Chapter 25

# Higher-codimension inversion and formalization

### 25.1 Cusp and caustic classification

A one-parameter inverse typically meets folds. Two or more tunable auxiliary parameters make cusps unavoidable. For a scalar forward function  $F(p; \lambda_1, \lambda_2)$ , a cusp occurs when

$$F_p = F_{pp} = 0, \quad F_{ppp} \neq 0,$$

with a transverse rank condition involving derivatives in the  $\lambda$  directions. The local normal form is

$$y - y_* = u^3 + \alpha u + \beta.$$

Its discriminant  $4\alpha^3 + 27\beta^2 = 0$  is the caustic separating regions with one and three real inverse branches.

The exceptional  $(a; q)_3$  slice at  $(-1/3, -1)$  is not itself a cusp in the full two-variable map; its Hessian is nondegenerate. It illustrates a different mechanism: a chosen parameter line can be tangent to a null cone of a Morse quadratic form and reveal a higher-order restricted singularity. This distinction should be enforced in any automated classifier.

**Research problem 25.1.** Compute the Thom–Boardman strata of the maps obtained by selecting two or three observables from finite Pochhammer or Gaussian families. Determine which stable singularities actually occur and express their loci in cyclotomic or partition-theoretic data.

### 25.2 Multivariate Lagrange–Good coefficients as combinatorial objects

Good’s determinant formula converts simultaneous recovery into coefficient extraction [7]. For  $q$ -analogs, the Jacobian determinant itself is assembled from Lambert sums and Gaussian polynomials. Its coefficients therefore have a second combinatorial interpretation superimposed on the usual rooted-tree interpretation of multivariate reversion.

**Conjecture 25.2** (**CONJECTURAL** Positivity after natural normalization). For simultaneous inverses formed from nested positive  $q$ -products, there is a canonical sign and monomial normalization after which the multivariate inverse coefficients are nonnegative integers counting decorated forests.

The conjecture should first be tested on the exact two-observable recovery  $((a; q)_n, (a; q)_{n+1})$ , then on adjacent Gaussian coefficients. A bijective proof would turn condition and identifiability determinants into enumerative objects.

## 25.3 A Lean formalization roadmap

The finite theory is particularly suitable for formal proof. A staged formalization can separate algebra, formal power series, and analysis.

**Stage 1: finite algebra.** Define  $(a; q)_n$  as a finite product and prove the argument-zero formula, recurrences, reciprocal transformations, and the discriminant factorization (20.11). Root-of-unity collision multiplicities reduce to counting residues in  $\mathbb{F}$  in  $n$ .

**Stage 2: formal reversion.** Use the existing formal-power-series inverse theorem to prove the Bell recurrence and instantiate it for finite products and Gaussian polynomials. For folds, adjoin a formal square root or work in Hahn/Puiseux series. The exact  $n = 3$  cubic can be proved by normalization and `ring`.

**Stage 3: real branch theorems.** Formalize coefficient positivity, strict monotonicity on  $q > 0$ , existence by the intermediate-value theorem, and uniqueness. This yields a fully certified positive Gaussian base inverse without complex analysis.

**Stage 4: analytic products.** Define the infinite product through summable logarithms, prove the tail bound (20.2), and obtain local inverses from the analytic implicit-function theorem. Endpoint asymptotics require a reusable Euler–Maclaurin library for functions with parameter-dependent singularities.

**Stage 5: certified computation.** Connect theorems to executable interval arithmetic. A certificate should contain a rational or dyadic box, a derivative enclosure, a residual bound, and a branch label; a small kernel theorem then validates the claimed inverse without trusting the search procedure.

## 25.4 Prioritized next results

The following sequence balances conceptual value against proof difficulty.

1. Prove and formalize the zero-fiber discriminant (20.11) and all finite root-collision multiplicities.
2. Compute secondary discriminants for  $n \leq 8$ , factor them, and formulate a stable pattern for cyclotomic and noncyclotomic factors.
3. Prove a uniform double-scaling expansion  $q = \zeta e^{-t}$ ,  $a = \zeta^{-r} e^{-\alpha t}$  for finite products, then pass to the infinite product.
4. Establish the complete real branch-count phase diagram for  $q \mapsto \Gamma_q(x)$  on a substantial interval of  $x$ , using interval certification where symbolic sign arguments are unavailable.
5. Derive and solve the differentiated refinement equation for the Fabius–Rvachev tangent CDF, and map its first critical curves in  $(q, x)$ .
6. Build a common branch-certificate format and use it for all numerical inverse claims in the `ProveIt`  $q$ -series drafts.

Each step produces reusable infrastructure for the next; none depends on an unverified global conjecture.

# Appendix A

## Coefficient machinery

### A.1 Recursive construction of a scalar inverse jet

Let

$$u = f(x) = \sum_{r \geq 1} a_r x^r, \quad a_1 \neq 0, \quad x = g(u) = \sum_{r \geq 1} b_r u^r.$$

Write  $[u^n]H(u)$  for coefficient extraction. Comparing coefficients in  $f(g(u)) = u$  gives

$$b_1 = \frac{1}{a_1}, \quad b_n = -\frac{1}{a_1} \sum_{k=2}^n a_k [u^n]g(u)^k \quad (n \geq 2), \quad (\text{A.1})$$

where only  $b_1, \dots, b_{n-1}$  occur on the right. Equivalently,

$$[u^n]g(u)^k = \frac{k!}{n!} B_{n,k}(1!b_1, 2!b_2, \dots),$$

with  $B_{n,k}$  the partial exponential Bell polynomial.

In derivative notation  $f_r = f^{(r)}(0)$ , the first five coefficients are

$$\begin{aligned} b_1 &= \frac{1}{f_1}, \\ b_2 &= -\frac{f_2}{2f_1^3}, \\ b_3 &= \frac{3f_2^2 - f_1 f_3}{6f_1^5}, \\ b_4 &= \frac{-15f_2^3 + 10f_1 f_2 f_3 - f_1^2 f_4}{24f_1^7}, \\ b_5 &= \frac{105f_2^4 - 105f_1 f_2^2 f_3 + 10f_1^2 f_3^2 + 15f_1^2 f_2 f_4 - f_1^3 f_5}{120f_1^9}. \end{aligned}$$

These universal rational polynomials are used repeatedly throughout the report. Computing forward derivatives logarithmically and applying (A.1) is generally simpler than expanding the original product before every inversion.

### A.2 Logarithmic forward derivatives

If  $F = e^L$ , then

$$F^{(n)} = F B_n(L', L'', \dots, L^{(n)}),$$

where  $B_n$  is the complete Bell polynomial. For a finite Pochhammer product, differentiation with respect to the argument gives

$$\partial_a^r \log(a; q)_n = -(r-1)! \sum_{j=0}^{n-1} \frac{q^{jr}}{(1-aq^j)^r}, \quad r \geq 1. \quad (\text{A.2})$$

At  $a = 0$ , the sum is  $[n]_{q^r}$ , producing all argument inverse coefficients from power sums of the geometric comb.

With  $q = e^t$ ,

$$\log(a; e^t)_n = \sum_{j=0}^{n-1} \log(1 - ae^{jt}),$$

and

$$\partial_t^r \log(a; e^t)_n \big|_{t=0} = -\text{Li}_{1-r}(a) \sum_{j=0}^{n-1} j^r, \quad r \geq 1. \quad (\text{A.3})$$

Since  $\text{Li}_{1-r}(a)$  is rational in  $a$  for integer  $r \geq 1$ , the full  $q = 1$  inverse jet is generated by Bernoulli power sums and rational functions.

### A.3 Fold coefficients through the cubic correction

Let

$$y - y_* = Au^2 + Bu^3 + Cu^4 + O(u^5), \quad A \neq 0,$$

and set  $s = (y - y_*)^{1/2}$ . The two sheets have

$$u = \sigma \frac{s}{\sqrt{A}} - \frac{B}{2A^2} s^2 + \sigma \frac{5B^2 - 4AC}{8A^{7/2}} s^3 + O(s^4), \quad \sigma \in \{+1, -1\}. \quad (\text{A.4})$$

In derivative notation  $A = F''(p_*)/2$ ,  $B = F'''(p_*)/6$ , and  $C = F^{(4)}(p_*)/24$ . Formula (A.4) is convenient for initializing both real branches on the side where  $(y - y_*)/A > 0$ .

### A.4 Asymptotic reversion with a reciprocal logarithm

Suppose

$$X = \frac{A}{t} + B + Ct + Dt^2 + O(t^3), \quad A \neq 0,$$

and absorb  $B$  into  $\hat{X} = X - B$ . Seeking

$$t = \frac{A}{\hat{X}} + \frac{\alpha}{\hat{X}^3} + \frac{\beta}{\hat{X}^4} + O(\hat{X}^{-5})$$

gives

$$\alpha = CA^2, \quad \beta = DA^3. \quad (\text{A.5})$$

There is no  $\hat{X}^{-2}$  term after the constant has been centered. If the forward expansion also contains  $t \log t$ , the inverse acquires  $(\log X)/X^3$  terms; ordinary Laurent reversion is then insufficient.



## Appendix B

### Derivations at special bases

#### B.1 The finite Pochhammer critical point at $q = -1$

Put

$$L_n(a, q) = \log(a; q)_n.$$

At  $q = -1$ , separate even and odd indices. The first base derivative is

$$\partial_q L_n(a, -1) = -a \sum_{j=1}^{n-1} \frac{j(-1)^{j-1}}{1 - a(-1)^j}.$$

For  $n = 2m$ , the even indices contribute  $m(m-1)/(1-a)$ , while the odd indices contribute  $m^2/(1+a)$ , with the signs inherited from  $(-1)^{j-1}$ . Solving the resulting linear numerator gives

$$a_*(2m) = \frac{1}{2m-1}.$$

For  $n = 2m+1$ , the analogous sums give

$$a_*(2m+1) = -\frac{1}{2m+1}.$$

Substitution in the second logarithmic derivative yields

$$\partial_q^2 L_{2m}(a_*, -1) = -\frac{(m+1)(2m-1)^2}{12m(m-1)}, \quad (\text{B.1})$$

$$\partial_q^2 L_{2m+1}(a_*, -1) = -\frac{(m-1)(2m+1)^2}{12m(m+1)}. \quad (\text{B.2})$$

The even formula is nonzero for  $m \geq 2$ ; the odd formula is nonzero for  $m \geq 2$ . For  $m = 1$ , namely  $n = 3$ , it vanishes and direct multiplication gives the exact cubic identity.

#### B.2 The Morse determinant at the exceptional cubic

For odd  $n = 2m+1$ , evaluate the Hessian of  $L_n(a, q)$  at  $(a, q) = (-1/(2m+1), -1)$ . Straightforward finite-sum simplification gives

$$\det \text{Hess}_{a,q} L_n = -\frac{(2m+1)^4(m^2+m+1)}{48m^2(m+1)^2} < 0. \quad (\text{B.3})$$

Because multiplication by the nonzero critical value preserves nondegeneracy at a critical point, the Hessian of  $(a; q)_n$  is also indefinite. In particular, at  $n = 3$  the full singularity is Morse even though the fixed- $a$  restriction begins cubically.

### B.3 Radial root-of-unity decomposition

Let  $q = \zeta e^{-t}$ , with  $\zeta$  primitive of order  $m$ , and  $|a| < 1$ . Starting from

$$\log(a; q)_\infty = - \sum_{r \geq 1} \frac{a^r}{r(1 - q^r)},$$

split the sum according to  $m \mid r$ . If  $r = m\ell$ , then

$$\frac{1}{1 - e^{-m\ell t}} = \frac{1}{m\ell t} + \frac{1}{2} + \frac{m\ell t}{12} + O(t^3).$$

The divisible terms therefore contribute

$$-\frac{\text{Li}_2(a^m)}{m^2 t} + \frac{1}{2m} \log(1 - a^m) - \frac{a^m}{12(1 - a^m)} t + O(t^3).$$

For  $m \nmid r$ , Taylor expansion at  $t = 0$  gives

$$\frac{1}{1 - \zeta^r e^{-rt}} = \frac{1}{1 - \zeta^r} - \frac{r\zeta^r}{(1 - \zeta^r)^2} t + O(t^2).$$

Combining both classes yields

$$C_\zeta(a) = \frac{1}{2m} \log(1 - a^m) - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r}{r(1 - \zeta^r)}, \quad (\text{B.4})$$

$$\mathcal{D}_\zeta(a) = \frac{a^m}{12(1 - a^m)} - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r \zeta^r}{(1 - \zeta^r)^2}. \quad (\text{B.5})$$

The sums converge absolutely for  $|a| < 1$ . The inverse follows from (A.5).

### B.4 Gaussian cumulants at $q = 1$

For  $q = e^t$ , write

$$\log \frac{[n]_q}{\binom{n}{k}} = \sum_{j=1}^k \left[ \log \frac{e^{(n-k+j)t} - 1}{(n-k+j)t} - \log \frac{e^{jt} - 1}{jt} \right].$$

Using

$$\log \frac{e^x - 1}{x} = \frac{x}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} x^{2r}$$

gives

$$\log \frac{[n]_q}{\binom{n}{k}} = \frac{k(n-k)}{2} t + \sum_{r \geq 1} \frac{B_{2r} P_{2r}(n, k)}{2r(2r)!} t^{2r}. \quad (\text{B.6})$$

Thus every odd cumulant beyond the first vanishes. This is the infinitesimal form of palindromicity. Applying scalar reversion to (B.6) produces the complete inverse jet without expanding the Gaussian polynomial itself.

## B.5 The $q$ -gamma expansion near one

With  $q = e^{-t}$ , the  $q$ -gamma definition

$$\Gamma_q(x) = (1 - q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty}$$

combines two nonuniform Pochhammer expansions. Cancellation of the  $\pi^2/(6t)$ ,  $\log t$ , and constant terms gives

$$\log \frac{\Gamma_{e^{-t}}(x)}{\Gamma(x)} = -\frac{(x-1)(x-2)}{4}t + \frac{(x-1)(x-2)(2x+3)}{144}t^2 + O(t^3). \quad (\text{B.7})$$

Differentiating in  $x$  gives

$$\psi_{e^{-t}}(x) - \psi(x) = -\frac{2x-3}{4}t + \frac{6x^2-6x-5}{144}t^2 + O(t^3). \quad (\text{B.8})$$

At  $x = 3/2$ , the linear term in the  $q$ -digamma expansion vanishes; inversion of its base therefore begins with a square root of the target displacement.

# Appendix C

## Implementation notes

### C.1 Branch labels in data structures

A numerical return value should carry more than a complex number. A minimal inverse-branch record contains

1. the recovered parameter and an error estimate or enclosure;
2. the parameter coordinate used, such as  $t = \log q$ ;
3. a base point and continuation path identifier;
4. winding numbers for every logarithm that enters the forward function;
5. the nearest critical value and an estimate of  $1/|F_p|$ ;
6. a residual in both the original and logarithmic equations;
7. a status flag: asymptotic initializer, Newton-converged, interval certified, or formally proved.

Without these fields, two numerically equal approximations reached on different sheets are indistinguishable to downstream code.

### C.2 Precision management

Near a simple critical point, the inverse condition behaves like  $|y - y_*|^{-1/2}$ ; near an order- $m$  point, like  $|y - y_*|^{-(m-1)/m}$ . The working precision should therefore be raised before the target enters the critical neighborhood. For a product evaluated as a logarithmic sum, add guard digits for cancellation among summands and for exponentiating the final logarithm.

A practical adaptive rule is

$$P_{\text{work}} = P_{\text{goal}} + \lceil -\log_{10} |F_p| \rceil + P_{\text{cancel}} + P_{\text{safety}},$$

where  $P_{\text{cancel}}$  is estimated from the ratio between the sum of absolute term magnitudes and the final logarithmic derivative. At a known Puiseux point, replace  $F_p$  by the derivative in the regularized root coordinate.

### C.3 Regression tests

A serious implementation should include at least the following tests.

1. Symbolic composition of every generated inverse jet through its claimed order.
2. Reciprocal round trips  $q \mapsto 1/q$  for palindromic polynomials.

3. Exact special values at  $q = 0, 1, -1$ , including all parity cases.
4. Monodromy tests around one simple fold and the exact cubic.
5. Tail-bound tests for infinite products against a doubled truncation.
6. Agreement between direct, logarithmic, and recurrence evaluators in overlapping safe regions.
7. Independent recovery of the atomic-family parameter from variance and kurtosis.
8. Interval certificates for a representative solution on every real branch class.

Tests should compare residual orders, not merely decimal values at one point.

## C.4 Files included in the archive

The archive contains the standalone LaTeX source, the rendered PDF, the Python experiment program, dependency list, generated CSV data, vector and raster figures, numerical summaries, and a SHA-256 manifest. Build commands and a short map of the source drafts are recorded in `README.md`.

# Conclusion

Inverse  $q$ -analysis is best understood as the geometry of partial inverse branches rather than as a search for one universal closed form. The same small collection of mechanisms recurs throughout the subject: Bell-polynomial reversion at regular points; Newton–Puiseux series at folds and collisions; reciprocal symmetry at infinity; Bernoulli and polylogarithmic asymptotics at  $q = 1$ ; cyclotomic residue classes at roots of unity; and invariant coordinates for simultaneous parameter recovery.

The finite Pochhammer family makes this structure unusually explicit. Its argument zeros lie on a geometric comb, its zero-fiber discriminant factors completely, and its  $q = -1$  critical slices include both universal folds and an exact cubic restriction of a Morse saddle. The infinite product replaces finite algebraic coverings by logarithmic and transseries geometry, but radial root-of-unity inversion still has a universal reciprocal-log normal form. Gaussian coefficients add positivity, palindromicity, partition cumulants, and strong global real uniqueness. The other  $q$ -special functions inherit their inverse behavior from these products, with new degeneracy curves when leading endpoint coefficients vanish.

The Fabius–Rvachev connection is not cosmetic. The same geometric comb that defines  $q$ -Pochhammer products supplies the weights of a random series whose  $q = 1/2$  member is the Rvachev distribution. Its cumulants provide exact inverse observables, while its CDF poses a genuinely new branch-geometry problem governed by a differentiated refinement operator. This creates a concrete bridge from  $q$ -series inversion to atomic functions, parameter identification, and geometric-comb interpolation.

The most promising next advances are uniform double scaling at roots of unity, secondary discriminants and monodromy, global real phase diagrams for  $q$ -gamma and  $q$ -beta base inversion, and certified tangent-CDF computation. These problems admit a productive division of labor among symbolic algebra, complex asymptotics, interval numerics, and formal proof.

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